Dear Editor:

Thank you and the reviewers for inviting us to revise our research paper, "Collaborative Multivariate Time Series Forecasting via Variable-Tailored Inter-Temporal Graph and Adaptive-Smooth Frequency Fusion". We have diligently and comprehensively revised the manuscript according to all four reviewers' comments, while doing our best to adhere to the journal's recommendation of a 20-page limit. Attached with this letter is our point-by-point response file. Given that the submission system allows only a single PDF upload for the response, and to facilitate an efficient review for each reviewer, we have consolidated our responses into one document. The page distribution is as follows:

- Reviewer 1: Pages 2-6
- Reviewer 2: Pages 7-28
- Reviewer 3: Pages 29-33
- Reviewer 4: Pages 34-37

We believe the quality of the manuscript has been significantly improved after this revision. Thank you again for your and the reviewers' valuable time and professional guidance.

Sincerely,

Jierui Lei, Peng Peng, Haina Tang, Xudong Zhang, Fangzheng Chen, Wenjian Zhang

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July 22, 2025

To Reviewer 1：

Response 1:

Q1: Complexity of the Framework: While each module is well-motivated, the full pipeline (ASFF-MLP → VT-ITGraph → FDG-Conv → CI-MLP) is quite complex.

Re: Dear reviewer, we thank you for your appreciation of our proposed modul es and for this valuable feedback. We argue that our framework, while compre hensive, is designed with a clear and principled workflow.

Our core component is the **Frequency Domain Multivariate Block (FDMB)**, which integrates the Adaptive-Smooth Frequency Fusion MLP (ASFF-MLP), Variable-Tailored Inter-Temporal Graph (VT-ITGraph), and Frequency Dilated Group Convolution (FDG-Conv) to perform embedding learning on multivariate time series. In the revised manuscript, we have made this clearer by detailing the computational sequence of the FDMB in **Equations (17-20)**. Following this, we explicitly summarize the overall three-stage process of the VTITG-ASFFN model: **History-Embedding → Prediction → Future-Embedding**, providing a more structured explanation that goes beyond the Fig.2 graphical description.

4 Methodology

In this section, we will delve into our proposed framework VTITG-ASFFN and core component modules. The VTITG-ASFFN architecture in Fig. 2 , which consists of History-Embedding *(a)*, Prediction *(b)* and Future-Embedding *(c)*. We introduce Adaptive-Smooth Frequency Fusion MLP (ASFF-MLP), Variable-Tailored Inter-Temporal Graph (VT-ITGraph), and Frequency Dilated Group Convolution (FDG-Conv) to create stackable Frequency Domain Multivariate Block (FDMB), fully leveraging them for end-to-end embedding learning. Furthermore, we use CI-MLP for predicting future time series based on historical data. The VTITG-ASFFN operates through process: ***History-Embedding*** → ***Prediction*** → ***Future-Embedding***.

4.4 Model Formulation and Forward Calculation

The Frequency Domain Multivariate Block implements the sequential encapsulation of the modules presented in the previous section 4.1, 4.2, 4.3:

$$X^{(\text{ASFF-MLP})} = \text{ASFF-MLP}(X), \tag{17}$$

$$W^{(\text{VT-ITG})} = \text{VT-ITGraph}(X^{(\text{ASFF-MLP})}), \tag{18}$$

$$X^{(\text{VT-ITG})} = X^{(\text{ASFF-MLP})} \times \sigma(W^{(\text{VT-ITG})}) + X^{(\text{ASFF-MLP})}, \tag{19}$$

$$X^{(\text{FDG})} = \text{FDG-Conv}(X^{(\text{VT-ITG})}). \tag{20}$$

where $X^{(\text{ASFF-MLP})}$, $X^{(\text{VT-ITG})}$, $X^{(\text{FDG})}$ all $\in \mathcal{R}^{b\times c\times t}$. Finally for calculation, we regard the forward process of VTITG-ASFFN as an end-to-end learning in Fig. 2:

- *History Embedding:* Learn the historical spatio-temporal correlations through FDMB$(\cdot)$ stacking: $X^{(l)} = \text{FDMB}^{(l)}_{\text{his}}\left(X^{(l-1)}\right), l = 2, ..., L$. Where $X^{(l)}$ all $\in \mathcal{R}^{b\times c\times i}$.
- *Prediction:* Use the CI-MLP$(\cdot)$ to predict $Y \in \mathcal{R}^{b\times c\times o}$: $Y = \text{CI-MLP}(X^{(\text{L})})$.
- *Future Embedding:* Perform the same operation on Y as History Embedding to get the final MTSF result $\hat{X}$: $Y^{(l)} = \text{FDMB}^{(l)}_{\text{pred}}\left(Y^{(l-1)}\right), l = 2, ..., L$. Where $Y^{(l)}$ all $\in \mathcal{R}^{b\times c\times o}$, $\hat{X} = Y^{(L)}$ and $\text{FDMB}^{(l)}_{\text{his}}, \text{FDMB}^{(l)}_{\text{pred}}$ are parameter-independent.

Noted that the objective of the Future-Embedding is to minimize the distance between the predictive distribution and the true distribution due to the label autocorrelation and temporal distribution shift of future time series. Meanwhile we choose $L1$ loss for VTITG-ASFFN's training because of its robust to noises Han et al. (2024).

To further clarify, we wish to reiterate the design philosophy behind the workflow of both the FDMB and the overall VTITG-ASFFN architecture:

(1) **On the FDMB block:** Each component serves a distinct and necessary purpose in a logical sequence:

- **ASFF-MLP** first enhances the embedded representation of the time series along the **temporal dimension** through a synergistic fusion of local and global frequency components, guided by adaptive smoothing.
- **VT-ITGraph** then addresses **multivariate dependencies**. After learning customized adaptive graphs for Multi-Attribute (MAV) and Multi-Entity (MEV) variables, it transfers these correlations into temporal context weights via a Channels-Time Graph to dynamically capture diverse spatio-temporal dependencies.
- Finally, **FDG-Conv** learns a comprehensive spatio-temporal representation by **fusing multi-source signals and mining primary temporal patterns**.

In summary, the ASFF-MLP, VT-ITGraph, and FDG-Conv modules respectively fulfill the distinct objectives of temporal learning, multivariate dependency learning, and spatio-temporal representation integration. Each module is purposefully designed, and we have taken steps to manage the FDMB's complexity by using a small graph embedding dimension and avoiding high-dimensional embeddings for the time steps, ensuring its complexity scales logarithmically or linearly with the input sequence length and number of time series variables. Moreover, the FDMB is an intuitive, stackable, and plug-and-play module that preserves dimensionality. This means only the CI-MLP in the "Prediction" stage is responsible for mapping the historical look-back length to the future prediction length. The stackability of the FDMB is demonstrated in our hyperparameter experiments.

(2) **On the VTITG-ASFFN architecture:** We employ a simple end-to-end architecture: **"History-Embedding" → "Prediction" → "Future-Embedding"**.

- The "History-Embedding" stage performs frequency-domain multivariate learning.
- The "Prediction" stage executes the forecast from the historical length to the future length.
- Finally, the "Future-Embedding" stage refines the forecast by correcting the discrepancy between the predictive distribution and the true distribution.

Overall, the VTITG-ASFFN workflow is an intuitive and clear three-stage representation learning process, not an arbitrarily complex one. Furthermore, in **Discussion RQ6**, we have conducted comparative experiments on the resource consumption and efficiency of VTITG-ASFFN against SOTA models in ultra-long-term forecasting. These results strongly validate the effectiveness and efficiency of our model.

Table 5 Ultra-Long MTSF and Operational Efficiency in DSMT (512 input length). VTITG-ASFFN is efficient due to low embed dim VT-ITGraph and parallel FDG-Conv.

Horizons	512		1024		2048		3072		Average		Memory.avg	Speed.avg
Models	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	(MB)	(s/iter)
VTITG-ASFFN*	**0.199**	**0.125**	**0.258**	**0.191**	**0.307**	**0.249**	**0.332**	**0.281**	**0.274**	**0.211**	**886**	**0.0024**
VTITG-ASFFN	**0.202**	**0.129**	**0.259**	**0.191**	**0.307**	**0.249**	**0.334**	**0.282**	**0.276**	**0.213**	3090	**0.0067**
iTransformer	0.217	0.147	0.277	0.220	0.338	0.301	0.370	0.342	0.301	0.253	**1582**	**0.0036**
TimeMixer	0.216	0.142	0.282	**0.216**	**0.335**	**0.279**	0.357	**0.311**	0.298	0.237	2340	0.0110
ModernTCN	0.218	0.154	0.283	0.239	0.345	0.321	0.380	0.368	0.307	0.271	**1230**	0.0074
MSGNet	0.230	0.163	0.295	0.249	0.359	0.338	0.400	0.400	0.321	0.288	2952	0.0498
Time-LLM	**0.206**	**0.134**	**0.267**	**0.203**	**0.323**	**0.274**	**0.356**	0.314	**0.288**	**0.231**	16036	0.2135

Thank you for your careful consideration of our response!

Response 2:

Q2: Clarity of Exposition: Several parts of the paper, especially the methodology section, are dense and mathematically under-explained. The logical flow is hard to follow without repeated readings. Simplifying and restructuring the presentation would significantly improve accessibility.

Re: Dear Reviewer, thank you for your insightful feedback pointing out the need for more rigorous mathematical explanations in our methodology and for enhanced clarity in our presentation. Following your valuable suggestions, we have conducted comprehensive revisions and enhancements to the methodology section. We have removed redundant formulas and added necessary mathematical justifications to better clarify the design rationale behind VTITG-ASFFN. The following is a summary of the major additions and changes, which can be found in our latest revised manuscript. Thank you again for your careful consideration of our response:

(1) We now provide a **gradient-based perspective** to explain how the Adaptive-Masking method within the **Adaptive-Smooth Frequency Fusion MLP** enables the model to self-learn the stationarity information of the time series.

To analyze the adaptivity from the gradient perspective, the error-driven back-propagation term for the k-th frequency component can be simplified to $\frac{\partial X^{(S)}(t)}{\partial M_k} = \mathcal{F}\{X^{(\mathrm{Org})}\}_k \cdot \frac{1}{T} e^{i2\pi kt/T}$. Furthermore, since $\frac{\partial \mathcal{L}}{\partial M_k} \propto \frac{\partial X^{(S)}(t)}{\partial M_k}$, this implies that if a frequency component k in the input signal consistently leads to prediction error, a steep gradient will be generated causing M_k to approach zero. Conversely, if a frequency component is crucial for accurate prediction, the gradient will reinforce its magnitude in M to attenuate noise and amplify signal-bearing components selectively.

(2) We have reframed the introduction of the **Variable-Tailored Inter-Temporal Graph**, explaining the design significance of the time-varying graph for multi-attribute variables (MAV) and the shared graph for multi-entity variables (MEV) from the perspective of their **different underlying data generation assumptions.**

Adaptive Graph for MAV and MEV. Our modeling for "Multi-Attribute" and "Multi-Entity" variables is based on different underlying data generation assumptions. For MAV we assume that each variable is controlled by its own distinct latent process $\mathcal{Z}c$ (e.g., meteorological attributes) and its joint probability is factorized as $\prod_c P(X_c|\mathcal{Z}c, \theta_c) \cdot \Psi(X_1, \ldots, X_c; t)$. Where we employ a time-varying adaptive graph $G_{\text{adp}}^{\text{mav}} \in \mathbb{R}^{c \times c \times t}$ to explicitly model this dynamic interaction term $\Psi(X_1, \ldots, X_c; t)$. For MEV we assume that all variables are driven by a shared latent factor $\mathcal{Z}$ (e.g., regional traffic systems). Its joint probability distribution can be expressed as $\int P(\mathcal{Z}) \prod_c P(X_c|\mathcal{Z}, \theta_c) d\mathcal{Z}$. Based on this, we use a global self-adaptive graph $G_{\text{adp}}^{\text{mev}} \in \mathbb{R}^{c \times c}$ to capture these relatively stable relationships induced by the $\mathcal{Z}$.

(3) We have **added a new design for the Variable-Tailored Inter-Temporal Graph to handle mixed MAV-MEV scenarios**. Furthermore, we have introduced a custom synthetic dataset in a new **Discussion RQ2** to demonstrate that VTITG-ASFFN can also effectively handle mixed scenarios where the distinction between MAV and MEV features is ambiguous. We argue that this is a valuable extension, even though variables in current mainstream benchmarks can often be clearly categorized as strictly MAV or MEV.

- *MAV-MEV Mixing:* Notably, while mainstream MTSF benchmarks can be categorized into attribute-dominant and entity-dominant types, hybrid scenarios with ambiguous feature representations commonly exist. To address this, we integrate a time-varying graph $G_{\text{adp}}^{\text{mav}} \in \mathbb{R}^{c \times c \times t}$ and a global enhanced graph $G_{\text{adp}}^{\text{mev}} \in \mathbb{R}^{c \times c}$ via adaptive gating fusion, where a dominance coefficient $\alpha_c \in \mathbb{R}^{c \times 1 \times 1}$ is learned to indicate feature tendency: $\alpha_c \to 1$ favors attribute features whereas $\alpha_c \to 0$ emphasizes entity characteristics. The resultant hybrid graph $G_{\text{adp}}^{\text{mix}} \in \mathbb{R}^{c \times c \times t}$ effectively handles mixed attributes and entities feature scenarios in discussion 6.2:

$$G_{\text{adp}}^{\text{mix}} = \text{Sigmoid}(\alpha_c) \cdot G_{\text{adp}}^{\text{mav}} + (1 - \text{Sigmoid}(\alpha_c)) \cdot G_{\text{adp}}^{\text{mev}} \tag{10}$$

here $\text{Sigmoid}(\cdot)$ is used to constrain the gating weights, while α_c will automatically find the optimal "identity" for each time series variable as the model minimizes the prediction loss. $G_{\text{adp}}^{\text{mev}}$ by broadcasting on the time-axis "t" to match $G_{\text{adp}}^{\text{mav}}$.

(4) We have added a description of the design purpose for the **Future-Embedding** stage, clarifying that its objective is to **minimize the discrepancy between the predictive distribution and the true distribution**.

- *Future Embedding:* Perform the same operation on Y as History Embedding to get the final MTSF result $\hat{X}$: $Y^{(l)} = \text{FDMB}^{(l)}\left(Y^{(l-1)}\right), l = 2, ..., L$. Where $Y^{(l)}$ all $\in \mathcal{R}^{b \times c \times o}$, $\hat{X} = Y^{(L)}$. Noted that the objective of the Future-Embedding is to minimize the distance between the predictive distribution and the true distribution due to the label auto-correlation and temporal distribution shift of future time series.

(5) We have **streamlined the Preliminary section by removing redundant theorems**, including Parseval's theorem and the autocorrelation formula, as they did not play a pivotal role in the methodology. We have retained five key points: Problem

Definition, Self-Adaptive Graph, Channels-Independent MLP, Globality in Discrete Fourier Transform, and Error-Driven Adaptive Smoothing. We believe that incorporating these points directly into the methodology would be slightly redundant, and presenting them as prior knowledge helps the reader build a more comprehensive understanding of the design principles of VTITG-ASFFN.

(6) We have **restructured the Related Work section** to be presented in three subsections: Multivariate Time Series Forecasting, Self-Adaptive Graph Modeling, and Frequency Domainization. The limitations of the latest research discussed in these subsections now **correspond directly to Challenge 1, Challenge 2, and Challenge 3** from the Introduction. This helps the reader better understand the challenges in MTSF and connects the innovations of VTITG-ASFFN to the difficulties of real-world tasks.

(7) We have provided a high-level overview before the main methodology description, after which we delve into the details of each component module of VTITG-ASFFN, and finally, we introduce the model's complete forward computation process. In this way, we have optimized the description of our methodology using a "general-to-specific-to-general" structure.

(8) We have aligned the Research Questions (RQs) in the Discussion section to correspond sequentially with Challenges I, II, and III. Furthermore, we have explicitly indicated in the subtitles which Challenge each RQ serves to address, creating a stronger contextual correspondence throughout the paper.

To Reviewer 2:

Code

Response 1:

Q1: I was not able to access the code, but it seems that the authors will provide it.

Re: Dear Reviewer, thank you for your valuable feedback and your positive assessment of our research results. We sincerely apologize for the oversight in not providing access to the code. We had initially omitted an anonymous link, considering the single-blind nature of the ACML journal track. To resolve this, we have now included the direct link to the original GitHub repository in our revised manuscript: https://github.com/leijieruilq/VTITG-ASFFN. We confirm that the full experimental framework code for VTITG-ASFFN will be made public. Thank you for the important reminder; this was our oversight.

Experimental section

Response 2:

Q2: The decision to only show the average over the 4 forecasting steps is difficult to understand (possibly lack of space?)

Re: Dear Reviewer, thank you for your recognition and encouragement of our research results. You are correct in your assessment; due to the 20-page limit of the main paper, we did not have sufficient space to present the performance for every forecast horizon individually. For this reason, we opted to show the average MAE and MSE results across the {96,192,336,720} horizons. We are certainly willing to include the full results in an appendix in future versions of the manuscript should space permit. For your convenience in this review process, we will now provide the **full results** of our comparative experiments, showing the performance of VTITG-ASFFN against all 12 baselines on both multi-attribute and multi-entity tasks at each specific forecast horizon. We believe this will help you to better assess the superior performance of VTITG-ASFFN at a more granular level compared to statistical, machine learning, deep learning, and large language models. Upon observing the full results on MAV and MEV tasks, one can see that while VTITG-ASFFN may not be strictly the top-performing model at every single forecast horizon, its comprehensive performance across all horizons and multiple datasets is consistent with the analysis in our manuscript and surpasses the SOTA baselines. In addition, a recent study (Lorenzo Brigato, et al., "Position: There are no Champions in Long-Term Time Series Forecasting") has shown that no single SOTA model consistently outperforms other baselines across all datasets and prediction ranges. Therefore, we chose to focus on the average performance on the MAV and MEV tasks to reflect the model's improvement in overall **consistency** and robustness.

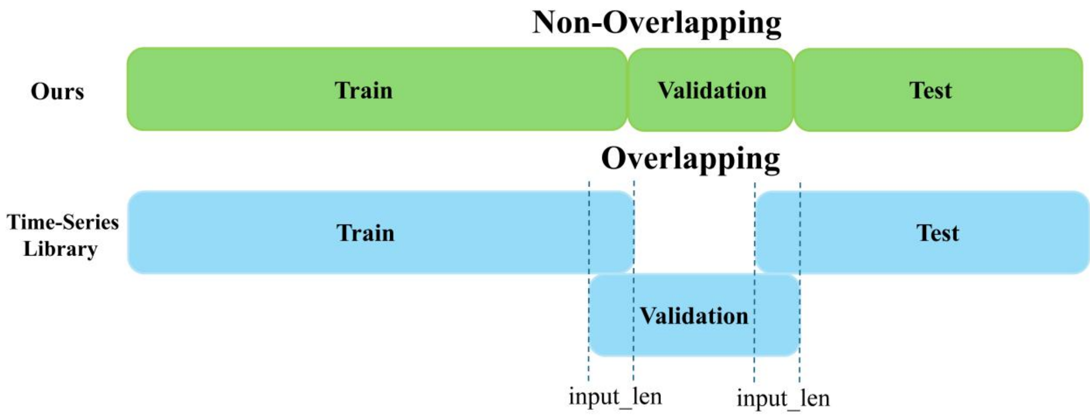
Dataset (Type)	Method	96		192		336		720		Average	
		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
ETTm1 (MAV)	ARIMA	0.666	0.983	0.686	1.019	0.706	1.055	0.733	1.099	0.698	1.039
	GBRT	0.382	0.372	0.408	0.416	0.437	0.457	0.476	0.520	0.426	0.441
	LSTM	0.534	0.640	0.559	0.686	0.583	0.733	0.615	0.790	0.573	0.712
	DLinear	0.372	0.345	0.389	0.380	0.413	0.413	0.453	0.474	0.407	0.403
	PatchTST	0.367	0.329	0.385	0.367	0.410	0.399	0.439	**0.454**	0.400	0.387
	TimesNet	0.375	0.338	0.387	0.371	0.411	0.410	0.450	0.478	0.406	0.399
	FilterNet	0.361	0.321	0.387	0.367	0.409	0.401	0.448	0.477	0.401	0.392
	MSGNet	0.366	0.319	0.397	0.376	0.422	0.417	0.458	0.481	0.411	0.398
	ModernTCN	0.365	0.322	0.394	0.372	0.415	0.406	0.452	0.473	0.407	0.393
	Timemixer	0.350	0.326	**0.372**	0.374	**0.396**	0.403	**0.432**	0.472	0.388	0.394
	iTransformer	0.368	0.334	0.391	0.377	0.420	0.426	0.459	0.491	0.410	0.407
	Time-LLM	0.372	0.330	0.394	0.371	0.409	0.398	0.440	**0.454**	0.403	0.388
	VTITG-ASFFN	**0.349**	**0.319**	**0.372**	**0.364**	**0.396**	**0.393**	0.434	0.455	**0.388**	**0.382**
ETTm2 (MAV)	ARIMA	0.270	0.137	0.292	0.161	0.316	0.189	0.352	0.235	0.308	0.180
	GBRT	0.238	0.129	0.269	0.164	0.301	0.202	0.346	0.253	0.289	0.187
	LSTM	0.426	0.363	0.438	0.368	0.487	0.435	0.530	0.492	0.470	0.414
	DLinear	0.234	0.120	0.258	0.148	0.293	0.187	0.347	0.224	0.283	0.170
	PatchTST	0.241	0.125	0.281	0.161	0.304	0.188	0.335	0.215	0.290	0.172
	TimesNet	0.232	0.123	0.261	0.155	0.287	0.186	0.319	0.224	0.274	0.172
	FilterNet	0.225	0.117	0.253	0.145	0.282	0.179	0.322	0.232	0.270	0.168
	MSGNet	0.231	0.121	0.262	0.154	0.294	0.198	0.340	0.261	0.282	0.184
	ModernTCN	0.228	0.120	0.262	0.156	0.281	0.183	0.325	0.239	0.274	0.175
	Timemixer	0.222	0.115	0.252	0.145	0.283	0.179	0.321	0.230	0.270	0.167
	iTransformer	0.248	0.133	0.268	0.156	0.295	0.190	0.333	0.243	0.286	0.180
	Time-LLM	0.228	0.119	0.255	0.146	0.283	0.180	0.323	0.232	0.272	0.169
	VTITG-ASFFN	**0.218**	**0.111**	**0.250**	**0.142**	**0.277**	**0.173**	**0.314**	**0.219**	**0.264**	**0.161**
DSMT-D1 (MAV)	ARIMA	0.100	0.028	0.159	0.070	0.221	0.130	0.305	0.229	0.196	0.114
	GBRT	0.096	0.029	0.170	0.078	0.242	0.142	0.352	0.258	0.215	0.127
	LSTM	0.352	0.225	0.390	0.258	0.435	0.306	0.547	0.453	0.431	0.310
	DLinear	0.234	0.120	0.258	0.148	0.293	0.187	0.347	0.224	0.283	0.170
	PatchTST	0.088	0.025	0.164	0.074	0.251	0.170	0.312	0.209	0.203	0.120
	TimesNet	0.079	0.019	0.149	0.062	0.208	0.121	0.291	0.213	0.181	0.103
	FilterNet	0.073	0.024	0.136	0.072	0.202	0.139	0.288	0.244	0.175	0.120
	MSGNet	0.078	0.024	0.146	0.079	0.212	0.152	0.306	0.274	0.185	0.132
	ModernTCN	**0.060**	0.018	0.128	0.070	0.201	0.145	0.291	0.251	0.170	0.121
	Timemixer	0.073	0.024	0.137	0.075	0.202	0.139	0.284	0.240	0.174	0.119
	iTransformer	0.079	0.026	0.148	0.079	0.212	0.122	0.303	0.226	0.185	0.113
	Time-LLM	0.080	0.025	0.142	0.073	0.208	0.138	0.296	0.244	0.181	0.120
	VTITG-ASFFN	0.061	**0.017**	**0.121**	**0.056**	**0.180**	**0.104**	**0.266**	**0.194**	**0.157**	**0.092**
Weather (MAV)	ARIMA	0.244	0.184	0.282	0.227	0.319	0.272	0.367	0.338	0.303	0.255
	GBRT	0.227	0.181	0.268	0.224	0.309	0.275	0.361	0.346	0.291	0.257
	LSTM	0.318	0.281	0.344	0.313	0.369	0.347	0.402	0.397	0.358	0.335
	DLinear	0.255	0.196	0.296	0.237	0.335	0.283	0.381	0.345	0.317	0.265
	PatchTST	0.206	0.156	0.248	0.199	0.291	0.249	0.343	0.336	0.272	0.235
	TimesNet	0.200	0.149	0.245	0.196	0.277	0.238	0.334	0.314	0.264	0.224
	FilterNet	0.207	0.162	0.250	0.210	0.290	0.265	0.340	0.342	0.272	0.245
	MSGNet	0.212	0.163	0.254	0.212	0.299	0.272	0.348	0.350	0.278	0.249
	ModernTCN	0.189	0.151	0.236	0.201	0.276	0.255	0.328	0.324	0.257	0.233
	Timemixer	0.200	0.166	0.241	0.211	0.279	0.259	0.329	0.331	0.262	0.242
	iTransformer	0.214	0.174	0.254	0.221	0.296	0.278	0.347	0.358	0.277	0.257
	Time-LLM	0.216	0.184	0.254	0.228	0.294	0.285	0.341	0.355	0.276	0.263
	VTITG-ASFFN	**0.177**	**0.141**	**0.226**	**0.186**	**0.268**	**0.232**	**0.308**	**0.291**	**0.245**	**0.213**

Dataset (Type)	Method	96		192		336		720		Average	
		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
QPS (MEV)	ARIMA	0.263	0.125	0.433	0.308	0.631	0.613	0.961	1.300	0.572	0.587
	GBRT	0.157	0.066	0.254	0.160	0.364	0.322	0.487	0.530	0.316	0.270
	LSTM	0.314	0.214	0.375	0.281	0.450	0.382	0.543	0.585	0.421	0.365
	DLinear	0.148	0.057	0.229	0.113	0.290	0.169	0.415	0.359	0.271	0.174
	PatchTST	0.180	0.076	0.238	0.118	0.309	0.163	0.384	0.260	0.278	0.154
	TimesNet	0.126	0.048	0.246	0.161	0.384	0.340	0.549	0.564	0.326	0.279
	FilterNet	0.119	0.044	0.198	0.111	0.295	0.220	0.496	0.534	0.277	0.227
	MSGNet	0.092	0.036	0.138	0.090	0.192	0.192	**0.253**	0.252	**0.169**	0.142
	ModernTCN	**0.081**	**0.027**	**0.132**	0.067	**0.185**	0.123	0.284	0.245	0.171	0.115
	Timemixer	0.127	0.050	0.242	0.161	0.328	0.266	0.437	0.453	0.284	0.232
	iTransformer	0.115	0.044	0.186	0.108	0.288	0.246	0.473	0.612	0.266	0.253
	Time-LLM	0.150	0.063	0.290	0.218	0.441	0.470	0.734	1.038	0.404	0.447
	VTITG-ASFFN	0.092	0.028	0.151	**0.063**	0.207	**0.095**	0.306	**0.200**	0.189	**0.097**
Traffic (MEV)	ARIMA	0.824	1.474	0.825	1.473	0.826	1.485	0.829	1.476	0.826	1.477
	GBRT	0.286	0.550	0.285	0.557	0.292	0.573	-	-	0.286	0.554
	LSTM	0.449	0.873	0.436	0.865	0.440	0.886	0.456	0.915	0.445	0.885
	DLinear	0.396	0.650	0.370	0.598	0.373	0.605	0.394	0.645	0.383	0.625
	PatchTST	0.319	0.583	0.331	0.591	0.332	0.599	0.341	0.601	0.331	0.594
	TimesNet	0.321	0.593	0.336	0.617	0.336	0.629	0.350	0.640	0.336	0.620
	FilterNet	0.298	0.462	0.302	0.472	0.308	0.487	0.327	0.523	0.309	0.486
	MSGNet	0.331	0.636	0.332	0.645	0.343	0.651	0.362	0.697	0.342	0.657
	ModernTCN	0.327	0.477	0.328	0.487	0.328	0.494	0.356	0.531	0.335	0.497
	Timemixer	0.285	0.462	0.296	0.473	0.296	0.498	0.313	0.506	0.298	0.485
	iTransformer	0.283	**0.428**	0.289	**0.447**	0.297	**0.462**	0.316	**0.499**	0.296	**0.459**
	Time-LLM	0.305	0.515	0.303	0.510	0.310	0.522	0.326	0.556	0.311	0.526
	VTITG-ASFFN	**0.271**	0.475	**0.277**	0.480	**0.282**	0.491	**0.303**	0.537	**0.283**	0.495
Electricity (MEV)	ARIMA	0.766	0.936	0.767	0.941	0.770	0.941	0.779	0.954	0.771	0.943
	GBRT	0.256	0.171	0.264	0.177	0.280	0.192	0.312	0.228	0.267	0.180
	LSTM	0.397	0.333	0.402	0.345	0.417	0.366	0.434	0.385	0.413	0.357
	DLinear	0.282	0.197	0.285	0.196	0.301	0.209	0.333	0.245	0.300	0.212
	PatchTST	0.268	0.159	0.278	0.177	0.296	0.195	0.317	0.215	0.290	0.187
	TimesNet	0.272	0.168	0.289	0.184	0.300	0.198	0.320	0.220	0.295	0.193
	FilterNet	0.260	0.182	0.267	0.187	0.282	0.203	0.314	0.243	0.281	0.204
	MSGNet	0.274	0.165	0.292	0.184	0.302	0.195	0.332	0.231	0.300	0.194
	ModernTCN	0.261	0.155	0.279	0.175	0.296	0.194	0.319	0.229	0.289	0.188
	Timemixer	0.247	0.153	0.256	0.166	0.277	0.185	0.310	0.225	0.273	0.182
	iTransformer	0.240	0.148	0.253	0.162	0.269	0.178	0.317	0.225	0.270	0.178
	Time-LLM	0.262	0.183	0.276	0.194	0.294	0.211	0.323	0.250	0.289	0.210
	VTITG-ASFFN	**0.230**	**0.135**	**0.250**	**0.154**	**0.264**	**0.170**	**0.294**	**0.205**	**0.259**	**0.165**
SoLar (MEV)	ARIMA	0.753	1.057	0.762	1.072	0.763	1.070	0.772	1.073	0.762	1.068
	GBRT	0.215	0.179	0.223	**0.189**	0.227	**0.194**	0.231	**0.201**	0.224	**0.191**
	LSTM	0.234	0.199	0.245	0.214	0.249	0.220	0.249	0.220	0.244	0.213
	DLinear	0.215	0.224	0.253	0.294	0.269	0.325	0.267	0.321	0.251	0.291
	PatchTST	0.194	0.256	0.223	0.271	0.231	0.275	0.236	0.276	0.221	0.270
	TimesNet	0.233	0.222	0.256	0.238	0.285	0.293	0.278	0.282	0.263	0.259
	FilterNet	0.218	0.185	0.245	0.232	0.252	0.245	0.253	0.245	0.242	0.227
	MSGNet	0.220	0.200	0.257	0.237	0.277	0.273	0.275	0.272	0.257	0.245
	ModernTCN	0.215	0.184	0.247	0.220	0.253	0.231	0.251	0.235	0.241	0.218
	Timemixer	0.227	0.207	0.254	0.257	0.251	0.263	0.265	0.279	0.249	0.251
	iTransformer	0.216	0.186	0.248	0.222	0.254	0.239	0.248	0.236	0.242	0.221
	Time-LLM	0.217	0.185	0.246	0.228	0.251	0.238	0.250	0.241	0.241	0.223
	VTITG-ASFFN	**0.194**	**0.177**	**0.217**	0.215	**0.225**	0.229	**0.219**	0.212	**0.213**	0.208

Response 3:

Q3: The result of Traffic for TimeMixer has MAE of 0.263 according to the authors of TimeMixer, while in your case it is reported 0.298. This is possibly explained by the difference in train splits, and by possibly different hyperparametrizations.

Re: Dear Reviewer, thank you for pointing out the discrepancy between our experimental results and those in the original paper. However, we believe there are several points worth clarifying. First, the result for TimeMixer in the original paper is an **MAE of 0.297**, not 0.263; we invite you to re-examine the publication. Even for iTransformer, one of the top-performing models on the Traffic dataset within the Time-Series Library framework, the average MAE is 0.282. You have correctly identified that our experimental framework can yield results that differ from papers based on the Time-Series Library. This is indeed a significant challenge in the field of multivariate time series forecasting, which we would like to elaborate on from the following perspectives:



(Datasets division for train, validation and test sets.)

(1) **On Dataset Splitting:** Time series forecasting is based on a sliding-window approach, where a look-back window of historical data is used to predict a future horizon. This means the training, validation, and test sets are derived by extracting samples from the original time series segments. However, the extraction method used in many existing libraries is overlapping. **For overlapping time series libraries, since the initial predictions partially rely on "known" historical data, this strategy may inadequately assess the model's real-time adaptability to sudden distribution shifts post-training, potentially yielding over-optimistic evaluations. Moreover, key hyperparameters (e.g., seq_len) tuned this way may overfit transitional patterns between training/validation periods rather than reflect generalizable performance.** Therefore, unlike studies that use the Time-Series Library, our "custom" data loader employs a **non-overlapping strategy**. This design ensures two things: 1) The input and label for each sample are generated completely without padding operations; 2) All samples originate exclusively from their

designated time interval (train, validation, or test), which prevents data leakage. As a result, the sample size and baseline performance may differ slightly from the original papers. We have mentioned the main points of this approach in the experimental setup section of our manuscript.

(2) **On Deep Learning Model Training:** The Time-Series Library provides functions for adaptive learning rate scheduling, and different scheduling strategies can lead to unfair comparisons. We have also noted that while most papers use a uniform look-back window size for comparison, this does not guarantee a completely fair comparison either. Some models may require more epochs to converge (e.g., ModernTCN), while others might be trained with different initial learning rates, batchsize (e.g., patchtst) or loss functions. Furthermore, much research tends to rely on various implementation tricks to boost performance, which can lead to hyperparameter overfitting and mask the true benefits of the proposed innovations. Based on this, we adopted a **unified training strategy** for all models to ensure fairness: all training experiments were configured with the **L1 loss** function and the **ADAM** optimizer. We used the same **96-step look-back window** to perform multi-step forecasting for horizons of 96, 192, 336, and 720. The maximum early stopping epoch was set to 42, with a learning rate of 1e-4 and a batch size of 32. It should be noted that for Time-LLM, we adjusted its batch size and text truncation length to fit within the 24GB GPU memory limit. Finally, we did not excessively search for hyperparameters for either VTITG-ASFFN or the baseline models, in order to reflect the **inherent differences** between these time series modeling approaches.

Presentation issues

Response 4:

Q4: The main issue of the paper is the difficulty to follow the main message over the different sections. I suggest the authors to extract the main challenges (in section I), and to map one to one each challenge to its related work (section II), its related component (section IV) and a discussion. In the current form, the section I includes 3 challenges, but the related work only include 2 unbalanced subsections, the methodology seems disconnected, while the research questions are only introduced in the last section. The abstract should also follow the same order (currently, the first challenge is MAV/MEV while it is second in the abstract).

Re: Dear Reviewer, thank you for your attention to our research and for your valuable feedback. We have revised the manuscript according to your suggestions. We would like to report on and describe these changes in detail as follows:

(1) We have restructured the **Related Work section** into three subsections, where the limitations discussed in each now directly map to the challenges presented in the Introduction:

- **Subsection 1: Multivariate Time Series Forecasting** (maps to Challenge I: Variable Type Difference).
- **Subsection 2: Self Adaptive Graph Modeling** (maps to Challenge II: Dynamic Spatial-Temporal Dependencies Capture).
- **Subsection 3: Frequency Domainization** (maps to Challenge III: Ignoring Localization and Stationary). We have now established a one-to-one correspondence between the shortcomings of current research and the challenges we aim to address.

(2) In our **Discussion section**, in addition to meticulously validating the feasibility of VTITG-ASFFN, we also indirectly address the three challenges you mentioned. The mapping is as follows:

- **RQ2 (MAV-MEV Mixing Study)** further resolves and deepens Challenge I (Variable Type).
- **RQ3 (Interpretability Study)** responds to both Challenge II (Dynamic Dependencies) and Challenge I (Variable Type).
- **RQ4 (Learning Pattern)** addresses the **Localization** problem from Challenge III.
- **RQ5 (Adaptive-Smooth Study)** relates to the **Stationary** aspect of Challenge III.

We fully understand your intention of creating a direct correspondence between the Challenges and the Discussion. However, given the page limits and our desire to present a complete set of experiments in the main text, we have chosen to further clarify this complex correspondence here to resolve your concern. We truly appreciate that your goal is to help us present our research in the best possible way. Therefore, we have done our best to align the RQs with the Challenges in the **subtitles of the Discussion section**. For the Methodology section, we were concerned that forcing a challenge-based structure might disrupt the narrative flow of the model's introduction. As a compromise, we have now mapped the components (ASFF-MLP, VT-ITGraph, and FDG-Conv) to their corresponding challenges in **Fig. 2(e)**, in an effort to best meet your suggestion.

(3) Regarding the **abstract**, we have adjusted the order to ensure that the challenges and their corresponding solutions (Variable-Tailored Inter-Temporal Graph and Adaptive-Smooth Frequency Fusion) are presented sequentially.

Thank you once again for your detailed and constructive feedback on our paper!

Response 5:

Q5: Overall, the authors should remove some technical details (that are interesting, but because of the page limit, they should be in a supplementary material; also, assuming the authors are providing a documented code, interested reader can follow the details by reading the code). In particular, the Section 3 is too heavy, and this whole section should be integrated in Section 4.

Re: Dear Reviewer, thank you for your valuable suggestion. We agree that the technical details in our preliminary section (Section 3) may have been overly detailed and required streamlining and optimization. This is a reasonable point. For instance, the inclusion of Parseval's theorem and the autocorrelation formula could be considered redundant in the main text. Their original purpose was to demonstrate that frequency-domain analysis is energetically equivalent to time-domain analysis (i.e., that our frequency-based learning does not cause information loss) and that autocorrelation can measure temporal dependencies. However, we acknowledge that these parts did not have a strong sequential connection with the methodology section. Therefore, we have removed them in the revised manuscript. However, we have retained five key points in Section 3 as essential preliminary details: **(1) Problem Definition, (2) Self-Adaptive Graph, (3) Channels-Independent MLP, (4) Globality in Discrete Fourier Transform, and (5) Error-Driven Adaptive Smoothing.** Our rationale for keeping these is as follows, and we invite you to re-examine Sections 3 and 4 of our revised manuscript:

- A clear **Problem Definition** is necessary to be established as a preliminary step for the task of multivariate time series forecasting.
- **Self-Adaptive Graph** is a common technique for modeling multivariate spatio-temporal dependencies. Our Variable-Tailored Inter-Temporal Graph is a two-stage dynamic graph learning method based on a customized adaptive graph. Therefore, it is necessary to introduce this concept beforehand; integrating it directly into the methodology section could create redundancy with our explanation of the Variable-Tailored Inter-Temporal Graph. This also serves as the foundation for how we capture the dynamic spatio-temporal dependencies mentioned in Challenge II.
- **Channels-Independent MLP**, as a linear layer technique primarily used in multivariate spatio-temporal forecasting, needs to be introduced as prior knowledge to help readers better understand how the Adaptive-Smooth Frequency Fusion MLP performs latent local information mining for frequency components. This also corresponds to our solution for the "ignoring localization" aspect of Challenge III.
- **Globality in Discrete Fourier Transform** explains the global nature of frequency components after a Fourier transform, providing justification for the problem of neglecting local information.
- **Error-Driven Adaptive Smoothing** is the data-driven solution to the "ignoring stationarity" aspect of Challenge III. Its purpose is to show that the design of the adaptive mask is analogous to a "window function" that is dynamically updated in both the time and frequency domains, which adaptively derives stationary information from a filtering perspective. Introducing it in the methodology section could lead to repetition, as we subsequently provide a gradient-based

analysis of the mask's "adaptive" nature, which logically follows from the introduction of this concept.

Although we have not merged the entirety of Section 3 into the methodology, we have retained the most comprehensive and essential preliminary knowledge. We believe this approach helps the reader to quickly grasp the fundamental technical details of MTSF and frequency-domain analysis, thereby leading to a better understanding of the methodology section. We sincerely thank you again for your recognition of our research and for your detailed questions!

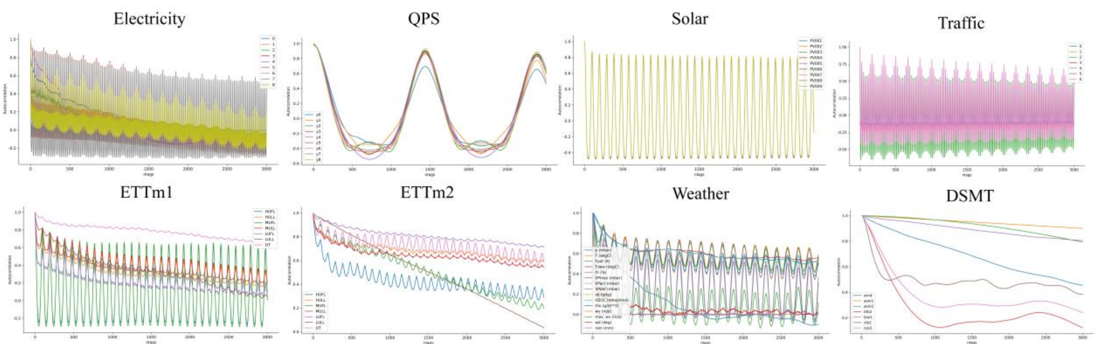
Response 6:

Q6: - Section 1: Each challenge could include a clear visualization with 1. the issue, and 2. the solution after using your proposed methodology. This would clarify and highlight each component of your methodology.

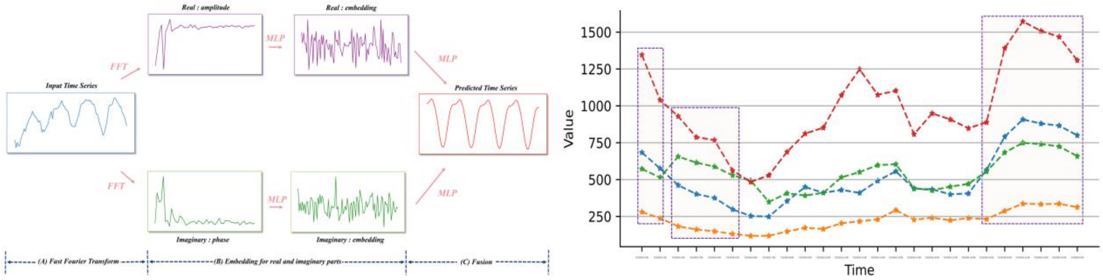
Re: Dear Reviewer, thank you for your valuable suggestion to incorporate visualizations into our Introduction (Section 1). Owing to page constraints, we have primarily focused on providing a corresponding set of analytical visualizations for the concept of 'Variable Tailoring' in Challenge I, while simplifying the illustrative diagrams for Challenges II and III. In the revised manuscript, within our discussion of Challenge II, we have now added a reference that points the reader to the corresponding spatio-temporal dependency analysis in Fig. 3.

Challenge II: **Dynamic Spatial-Temporal Dependencies Capture.** Self-adaptive graphs have the ability to capture multivariate relationships without prior knowledge. However, they are limited to learn the static graph representations, which fail to establish connections with temporal evolution. In other words, they overlook the critical aspect of "dynamic spatial-temporal dependencies" Yi et al. (2024b). We have observed the daily electricity usage data in a specific real-world region in Fig. 3 top. It was found that while the neighboring households tend to follow similar patterns Shao et al. (2022), which is also based on specific time periods. Therefore, multivariate dependencies are inherently diverse in temporal context.

Although we have not included a clear visual diagram for each Challenge in the revised manuscript, we are pleased to provide comprehensive visualizations for these challenges in this point-by-point response for your clarification:



(Challenge I: Auto-correlation within time-delayed variations. Top: "MEV" datasets; Bottom: "MAV" datasets. From observation,"MAV" reflects differences in temporal correlations while "MEV" are similar.)



(Challenge II, III: Left: Mining latent patterns and global fusion, which refines local granularity (from coarse to fine) and promotes modal complementarity from the perspectives of amplitude and phase; Right: The electricity usage reveals "dynamic spatial dependencies", i.e. similar patterns are manifested only in specific time periods)

Furthermore, we understand your preference for a one-to-one mapping where each of our methods individually solves a corresponding challenge. However, our approach is designed to be synergistic rather than a set of separate, one-to-one solutions. For example, our Variable-Tailored Inter-Temporal Graph simultaneously addresses Challenge I and II by providing customized adaptive graphs based on multi-attribute and multi-entity features, and then transfers these self-learned multivariate relationships into the temporal context to achieve dynamization. Therefore, our preferred narrative structure is to first identify the current research challenges, and then to elaborate on our method's innovations from the perspective of "jointly improving frequency domain learning and tailored dynamic graph modeling". Finally, in the main contributions, we summarize our proposed components from the two aspects of 'temporal learning' and 'multivariate correlations'. In this way, we aim to present the characteristics of VTITG-ASFFN through the lens of this synergistic and holistic approach. Finally, we would like to thank you once again for your detailed suggestions regarding the introduction section of our paper.

Response 7:

Q7: - Section 2: The section is unbalanced and difficult to read. The sentence at the end of each section could be explicitly illustrated on a specific example (not necessarily in Sec. 2), and connected to a challenge and a solution, with an illustrated effect of the same example. I'm referring to those 2 sentences: "failing to capture diverse spatio-temporal dependencies" and "lacking local-global collaborative learning for frequency components".

Re: Dear Reviewer, thank you for your valuable suggestions for revising our Related Work section (Section 2). In the revised manuscript, we have divided the Related Work section into three subsections: Multivariate Time Series Forecasting, Self Adaptive Graph Modeling, and Frequency Domainization. The limitations discussed in each of these subsections now correspond to Challenge I, II, and III from the Introduction (Section 1), respectively. Furthermore, in the Discussion section, we have also mapped our Research Questions (RQs) to the Challenges to demonstrate the connection between specific problems and our solutions, while also showcasing the practical effectiveness of our case studies. The following changes can be found in our revised manuscript:

2 Related Work

2.1 Multivariate Time Series Forecasting

Early time series forecasting primarily relied on traditional statistical models and classical machine learning approaches. The ARIMA model Shumway et al. (2017) integrates three components: autoregression (AR), differencing (I), and moving average (MA), generating forecasts based on historical time-window data. Gradient boosted regression trees (GBRT) Chen and Guestrin (2016); Han et al. (2024), by ensembling multiple weak learners with temporal feature engineering, demonstrated competitive performance in specific scenarios. However, such models face inherent limitations in efficiently parallelizing multivariate time-series forecasting, and their dominance has been challenged by the advancement of deep learning techniques. We focus on the advanced MTSF methods including MLP, TCN, Transformer, LLM and Graph based on deep learning. For MLP-based, DLinear Zeng et al. (2023) utilizes a simple linear layer to independently predict the seasonal and trend components of the variables; TimeMixer Wang et al. (2024) breaks down mixed time variations into multiple components (de-entanglement) to highlight the intrinsic properties of the time series. For TCN-based, ModernTCN Luo et al. (2024) jointly designs depthwise and grouped pointwise convolution to learn cross variable and temporal dependencies. For Transformer-based, PatchTST Nie et al. (2023) proposes patch modeling with channel independent approach and Transformer; iTransformer Liu et al. (2024) utilizes inverted dimensional attention to encode variable as token for representations learning. For LLM-based, Time-LLM Jin et al. (2024b) recodes time series through

cross-modal attention and uses domain knowledge to guide LLM inference. However, they mainly focus on temporal patterns. From graph structure learning perspective, although CrossGNN and MSGNet Huang et al. (2023); Cai et al. (2024) establish adaptive graph across different time scales, they heavily rely on manually defined scale extraction, making it challenging to capture "dynamic spatial-temporal dependencies". What's more, none of them try to take into account multivariate types (***Challenge I***).

2.2 Self Adaptive Graph Modeling

Adaptive graph is used to capture spatial dependencies in traffic flow. Graph WaveNet Wu et al. (2019) initializes the source and target embedding for each node and builds asymmetric matrix to represent the directed relationships. AGCRN Bai et al. (2020) provides only a single embedding for each node and computes the similarity. MTGNN Wu et al. (2020) uses the argtopk algorithm to dynamically select the most relevant k-node pairs to reduce the computational complexity. DMSTGCN Han et al. (2021) constructs dynamic adaptive matrix for multifaceted inter-node relationship capture and fusion. MegaCRN Jiang et al. (2023) learns spatio-temporal heterogeneity with Meta-Graphs. However, none of them have attempted to transfer the multivariate correlations of adaptive graph into temporal context for further temporal correlations mining, failing to capture diverse spatio-temporal dependencies (***Challenge II***).

2.3 Frequency Domainization

Existing works focus on fourier transform to appropriately represent time series. Autoformer Wu et al. (2021) designs an auto-correlation mechanism based on the Fourier transform to capture sequence periodicity for correlation discovery and representation aggregation at the subsequence level. Fedformer Zhou et al. (2022) combines seasonal and trend decomposition, Fourier analysis and Transformer to better capture global properties of time series. TimesNet Wu et al. (2022) inspired by multi-periodicity to capture intra- and inter-periodic variations in frequency domain. FilterNet Yi et al. (2024a) selectively passes components by learnable frequency filters. However, they seldom consider "adaptively" pre-smoothing the time series, meanwhile lacking the local-global collaborative learning for frequency components (***Challenge III***).

6.2 MAV-MEV Mixing Study (RQ2 for Challenge I)

6.3 Interpretability Study (RQ3 for Challenge I, II)

6.4 Learning Pattern and Prediction (RQ4 for Challenge III)

6.5 Adaptive-Smooth Study (RQ5 for Challenge III)

We also noticed your question regarding the intended meaning of two phrases: **"failing to capture diverse spatio-temporal dependencies"** and **"lacking local-global collaborative learning for frequency components".** We would like to clarify our perspective on these two points:

(1) Our understanding is that previous adaptive graph-based learning methods primarily learn a **static** multivariate graph structure from a dynamically evolving time series window. However, they do not attempt to transfer this learned graph structure into the temporal context to obtain contextual weights, akin to an "inter-temporal graph." The essential purpose of modeling multivariate dependencies is to aid in mining the temporal contextual correlations within the time series themselves. Therefore, our **Variable-Tailored Inter-Temporal Graph**, through its two-stage graph learning process, is able to capture these diverse spatio-temporal dependencies.

(2) Current mainstream frequency-domain learning methods used in multivariate time series (such as Autoformer, Fedformer, and FilterNet) either fully leverage the global properties of the frequency domain or integrate local information via subsequence auto-correlation aggregation. Both approaches lack an explicit mechanism for collaborative local-global dependency learning. In contrast, our VTITG-ASFFN utilizes the **Adaptive-Smooth Frequency Fusion MLP**, "realizing collaborative local-global learning and targeted smoothing guidance for diverse time series." To comprehensively integrate, we also introduce the **Frequency Dilated Group Convolution**, which "facilitates multi-source signal fusion and primary temporal patterns mining."

Furthermore, in **Discussion RQ4**, the **Learning Patterns Visualization** demonstrates the local-global consistent learning process of VTITG-ASFFN. Subsequently, the **Prediction Study** presents a comparison of the real-world prediction consistency of VTITG-ASFFN against other frequency domain-based MTSF models.

We hope this detailed analysis helps to clarify these two limitations. Due to overall page length constraints, we provided a more concise summary in the Related Work section of the main paper.

Response 8:

Q8: - Section 3: why do you need to introduce e.g. the Parseval's theorem? First, this theorem is not explicitly cited in the rest of the paper, and then, there are some weak assumptions for the theorem to be applicable. Are you using the notations S_h and S_f in Section 3.1? Is C in Section 3.1 equivalent to c in Section 4? I don't understand the sentence "therefore we mainly propose weak smoothness" at the end this section.

Re: Dear Reviewer, thank you for your valuable questions regarding our preliminary section (Section 3):

- As we also realized, Parseval's theorem did not play a pivotal role in the subsequent sections of the paper. Therefore, following your valuable suggestion, we have removed it from the revised manuscript.
- Regarding the symbols Sh and Sf, they are used in the problem definition for multivariate time series forecasting within Section 3. You can find these two symbols in the revised manuscript, where they appear as subscripts:

(1) Problem Definition. The Multivariate Time Series (MTS) Forecasting refers to predict future period based on historical time series in time windows form (i.e., look-back window): $\{X_{t+S_h}^{\{1,\ldots,c\}}, \ldots, X_{t+S_h+S_f-1}^{\{1,\ldots,c\}}\} = f_\theta(X_t^{\{1,\ldots,c\}}, \ldots, X_{t+S_h-1}^{\{1,\ldots,c\}})$. Where S_h and S_f are the time step length of historical time series and future time series. c is the number of time series variables. $f_\theta(\cdot)$ denotes MTSF models.

- Regarding the notation, C and c are equivalent and both refer to the number of multivariate time series variables. For the sake of consistency, we have changed all instances of the uppercase C to lowercase c throughout the manuscript. This change was made because lowercase letters are conventionally used to denote dimensional variables in formulas.
- We thank the reviewer for giving us the opportunity to clarify the concept of "weak smoothness". The "strictly smooth" time series we refer to is an idealized, mathematically perfect state, such as a perfect sine wave or a trend line without any noise or abrupt changes. However, real-world time series are inherently non-smooth, filled with noise, seasonality, and structural breaks. Forcing such a series to become strictly smooth risks **over-smoothing**, which would destroy vital predictive information (e.g., sudden market turning points or anomalous signals). Therefore, the **"weak smoothness"** we propose is a more pragmatic and balanced strategy. Its goal is **not** to create a perfectly smooth curve, but rather to **selectively** reduce the non-stationarity and high-frequency noise that are detrimental to frequency-domain analysis, while **preserving** the core dynamics of the original series.

Response 9:

Q9: - In the methodology section, the subsections are too technical and difficult to follow (list of formulas in Sec. 4.4, 4.5, too much text on tiny details in Sec. 4.1 to 4.3.

The high-level presentation should be at the beginning of the whole section not at the end, and should be non technical.

Re: Dear Reviewer, thank you very much for your valuable feedback regarding the excessive number of formulas and redundant descriptions in our methodology section. We have since thoroughly reorganized and revised this section by reducing the number of process-oriented formulas and incorporating more high-level, non-technical descriptive analysis. The following changes can be found in our revised manuscript:

(1) We now provide a **gradient-based perspective** to explain how the Adaptive-Masking method within the **Adaptive-Smooth Frequency Fusion MLP** enables the model to self-learn the stationarity information of the time series.

To analyze the adaptivity from the gradient perspective, the error-driven back-propagation term for the k-th frequency component can be simplified to $\frac{\partial X^{(S)}(t)}{\partial M_k} = \mathcal{F}\{X^{(\mathrm{Org})}\}_k \cdot \frac{1}{T} e^{i2\pi kt/T}$. Furthermore, since $\frac{\partial \mathcal{L}}{\partial M_k} \propto \frac{\partial X^{(S)}(t)}{\partial M_k}$, this implies that if a frequency component k in the input signal consistently leads to prediction error, a steep gradient will be generated causing M_k to approach zero. Conversely, if a frequency component is crucial for accurate prediction, the gradient will reinforce its magnitude in M to attenuate noise and amplify signal-bearing components selectively.

(2) We have reframed the introduction of the **Variable-Tailored Inter-Temporal Graph**, explaining the design significance of the time-varying graph for multi-attribute variables (MAV) and the shared graph for multi-entity variables (MEV) from the perspective of their **different underlying data generation assumptions.**

Adaptive Graph for MAV and MEV. Our modeling for "Multi-Attribute" and "Multi-Entity" variables is based on different underlying data generation assumptions. For MAV we assume that each variable is controlled by its own distinct latent process $\mathcal{Z}c$ (e.g., meteorological attributes) and its joint probability is factorized as $\prod_c P(X_c|\mathcal{Z}c, \theta_c) \cdot \Psi(X_1, \ldots, X_c; t)$. Where we employ a time-varying adaptive graph $G^{\mathrm{mav}}_{\mathrm{adp}} \in \mathbb{R}^{c \times c \times t}$ to explicitly model this dynamic interaction term $\Psi(X_1, \ldots, X_c; t)$. For MEV we assume that all variables are driven by a shared latent factor $\mathcal{Z}$ (e.g., regional traffic systems). Its joint probability distribution can be expressed as $\int P(\mathcal{Z}) \prod_c P(X_c|\mathcal{Z}, \theta_c) d\mathcal{Z}$. Based on this, we use a global self-adaptive graph $G^{\mathrm{mev}}_{\mathrm{adp}} \in \mathbb{R}^{c \times c}$ to capture these relatively stable relationships induced by the $\mathcal{Z}$.

(3) We have **added a new design for the Variable-Tailored Inter-Temporal Graph to handle mixed MAV-MEV scenarios**. Furthermore, we have introduced a custom synthetic dataset in a new **Discussion RQ2** to demonstrate that VTITG-ASFFN can also effectively handle mixed scenarios where the distinction between MAV and MEV features is ambiguous. We argue that this is a valuable extension, even though variables in current mainstream benchmarks can often be clearly categorized as strictly MAV or MEV.

• *MAV-MEV Mixing:* Notably, while mainstream MTSF benchmarks can be categorized into attribute-dominant and entity-dominant types, hybrid scenarios with ambiguous feature representations commonly exist. To address this, we integrate a time-varying graph $G_{\text{adp}}^{\text{mav}} \in \mathbb{R}^{c\times c\times t}$ and a global enhanced graph $G_{\text{adp}}^{\text{mev}} \in \mathbb{R}^{c\times c}$ via adaptive gating fusion, where a dominance coefficient $\alpha_c \in \mathbb{R}^{c\times 1\times 1}$ is learned to indicate feature tendency: $\alpha_c \to 1$ favors attribute features whereas $\alpha_c \to 0$ emphasizes entity characteristics. The resultant hybrid graph $G_{\text{adp}}^{\text{mix}} \in \mathbb{R}^{c\times c\times t}$ effectively handles mixed attributes and entities feature scenarios in discussion 6.2:

$$G_{\text{adp}}^{\text{mix}} = \text{Sigmoid}(\alpha_c) \cdot G_{\text{adp}}^{\text{mav}} + (1 - \text{Sigmoid}(\alpha_c)) \cdot G_{\text{adp}}^{\text{mev}} \tag{10}$$

here Sigmoid(·) is used to constrain the gating weights, while α_c will automatically find the optimal "identity" for each time series variable as the model minimizes the prediction loss. $G_{\text{adp}}^{\text{mev}}$ by broadcasting on the time-axis "t" to match $G_{\text{adp}}^{\text{mav}}$.

(4) We have added a description of the design purpose for the **Future-Embedding** stage, clarifying that its objective is to **minimize the discrepancy between the predictive distribution and the true distribution**.

• *Future Embedding:* Perform the same operation on Y as History Embedding to get the final MTSF result $\hat{X}$: $Y^{(l)} = \text{FDMB}^{(l)}\left(Y^{(l-1)}\right), l = 2, ..., L$. Where $Y^{(l)}$ all $\in \mathcal{R}^{b\times c\times o}$, $\hat{X} = Y^{(L)}$. Noted that the objective of the Future-Embedding is to minimize the distance between the predictive distribution and the true distribution due to the label auto-correlation and temporal distribution shift of future time series.

(5) In the revised Section 4.4, we have reorganized the description of the end-to-end execution flow of VTITG-ASFFN and have now explicitly defined the training loss function.

4.4 Model Formulation and Forward Calculation

The Frequency Domain Multivariate Block implements the sequential encapsulation of the modules presented in the previous section 4.1, 4.2, 4.3:

$$X^{(\text{ASFF-MLP})} = \text{ASFF-MLP}(X), \tag{17}$$

$$W^{(\text{VT-ITG})} = \text{VT-ITGraph}(X^{(\text{ASFF-MLP})}), \tag{18}$$

$$X^{(\text{VT-ITG})} = X^{(\text{ASFF-MLP})} \times \sigma(W^{(\text{VT-ITG})}) + X^{(\text{ASFF-MLP})}, \tag{19}$$

$$X^{(\text{FDG})} = \text{FDG-Conv}(X^{(\text{VT-ITG})}). \tag{20}$$

where $X^{(\text{ASFF-MLP})}$, $X^{(\text{VT-ITG})}$, $X^{(\text{FDG})}$ all $\in \mathcal{R}^{b\times c\times t}$. Finally for calculation, we regard the forward process of VTITG-ASFFN as an end-to-end learning in Fig. 2:

• *History Embedding:* Learn the historical spatio-temporal correlations through FDMB(·) stacking: $X^{(l)} = \text{FDMB}_{\text{his}}^{(l)}\left(X^{(l-1)}\right), l = 2, ..., L$. Where $X^{(l)}$ all $\in \mathcal{R}^{b\times c\times i}$.

• *Prediction:* Use the CI-MLP(·) to predict $Y \in \mathcal{R}^{b\times c\times o}$: $Y = \text{CI-MLP}(X^{(\text{L})})$.

• *Future Embedding:* Perform the same operation on Y as History Embedding to get the final MTSF result $\hat{X}$: $Y^{(l)} = \text{FDMB}_{\text{pred}}^{(l)}\left(Y^{(l-1)}\right), l = 2, ..., L$. Where $Y^{(l)}$ all $\in \mathcal{R}^{b\times c\times o}$, $\hat{X} = Y^{(L)}$ and $\text{FDMB}_{\text{his}}^{(l)}$, $\text{FDMB}_{\text{pred}}^{(l)}$ are parameter-independent.

Noted that the objective of the Future-Embedding is to minimize the distance between the predictive distribution and the true distribution due to the label auto-correlation and temporal distribution shift of future time series. Meanwhile we choose $L1$ loss for VTITG-ASFFN's training because of its robust to noises Han et al. (2024).

4 Methodology

In this section, we will delve into our proposed framework VTITG-ASFFN and core component modules. The VTITG-ASFFN architecture in Fig. 2 , which consists of History-Embedding *(a)*, Prediction *(b)* and Future-Embedding *(c)*. We introduce Adaptive-Smooth Frequency Fusion MLP (ASFF-MLP), Variable-Tailored Inter-Temporal Graph (VT-ITGraph), and Frequency Dilated Group Convolution (FDG-Conv) to create stackable Frequency Domain Multivariate Block (FDMB), fully leveraging them for end-to-end embedding learning. Furthermore, we use CI-MLP for predicting future time series based on historical data. The VTITG-ASFFN operates through process: ***History-Embedding*** $\rightarrow$ ***Prediction*** $\rightarrow$ ***Future-Embedding***.

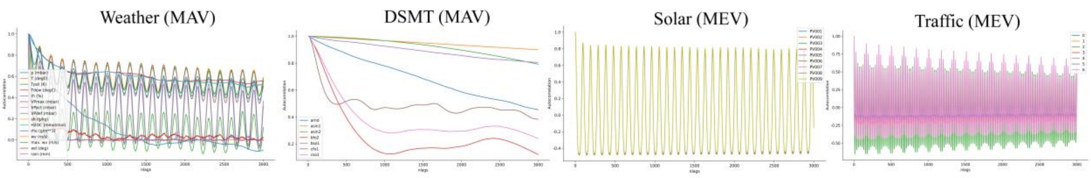
(6) At the beginning of the methodology section, we now provide a high-level overview of the overall VTITG-ASFFN architecture. This provides a top-down, summary introduction to the framework before we delve into the detailed analysis of each core component.

Figures

Response 10:

Q10: In Fig. 1, all eight datasets are put, but it is not necessary. The plot should include the mention of MEV and MAV for clarity. The axis are currently not readable while printed: I needed to zoom on the pdf version to understand that the plot is showing the auto-correlation. I don't understand "MEV are similar", e.g. for Electricity it does not seem to be the case.

Re: Dear Reviewer, thank you for your close attention to our figures. During the revision process, we came to agree that presenting all of the multi-attribute and multi-entity datasets in Fig. 1 was somewhat redundant. Therefore, we have re-selected and retained the temporal auto-correlation variations for four representative datasets: Solar (MEV), Traffic (MEV), Weather (MAV), and DSMT (MAV). The revised figure is shown below:



(Temporal auto-correlation variations in "MAV" and "MEV" scenarios)

Here we would like to clarify our statement that "MEV are similar". For multi-entity time series variables, the evolution of their autocorrelation values with increasing time lag is highly consistent, as seen in the Solar and Traffic datasets. This is because each entity variable essentially reflects the same underlying attribute. In contrast, multi-attribute variables reflect distinct physical properties (e.g., pressure, temperature) that have significant conceptual gaps between them. As a result, the evolution of their time-lagged autocorrelation varies greatly, as seen in the Weather and complex system (DSMT) datasets. To reiterate, multi-entity variables refer to records of the same observation type across different entities (e.g., traffic sensors). As these entities are conceptually similar and measure a uniform attribute, the essence of their time series evolution—the evolution of their time-lagged correlation—is also broadly similar. Even in a dataset like Electricity, where some entities might exhibit minor variations, their autocorrelations still fall within the overall distribution of the single, unified attribute being measured: electricity consumption. However, to better illustrate the essential difference between these variable types, we have revised the figure to remove the Electricity dataset and have instead selected four clearer examples—Solar (MEV), Traffic (MEV), Weather (MAV), and DSMT (MAV)—to more clearly showcase the "Temporal auto-correlation variations in 'MEV' and 'MAV' scenarios." This leads to a key insight into the difference in their temporal patterns: 'MAV' models should account for attribute-specific dynamics, while 'MEV' models can leverage shared patterns. This analysis of characteristics, derived from observation, helps justify our approach of modeling multivariate dependencies based on different data generation assumptions for multi-entity and multi-attribute variables:

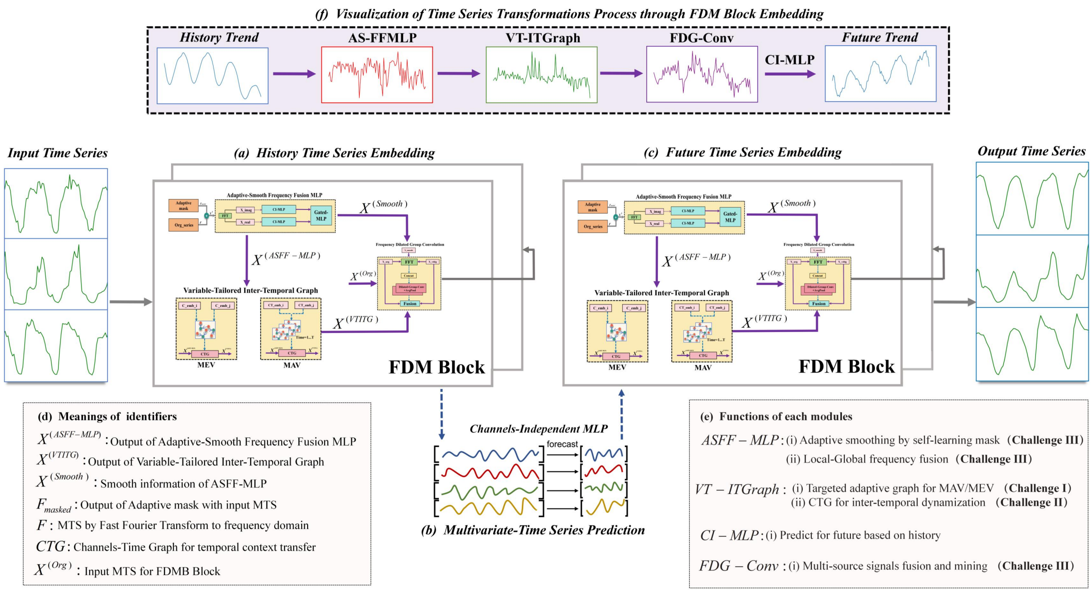
Adaptive Graph for MAV and MEV. Our modeling for "Multi-Attribute" and "Multi-Entity" variables is based on different underlying data generation assumptions. For MAV we assume that each variable is controlled by its own distinct latent process $\mathcal{Z}c$ (e.g., meteorological attributes) and its joint probability is factorized as $\prod_c P(X_c|\mathcal{Z}c, \theta_c) \cdot \Psi(X_1, \ldots, X_c; t)$. Where we employ a time-varying adaptive graph $G_{\text{adp}}^{\text{mav}} \in \mathbb{R}^{c \times c \times t}$ to explicitly model this dynamic interaction term $\Psi(X_1, \ldots, X_c; t)$. For MEV we assume that all variables are driven by a shared latent factor $\mathcal{Z}$ (e.g., regional traffic systems). Its joint probability distribution can be expressed as $\int P(\mathcal{Z}) \prod_c P(X_c|\mathcal{Z}, \theta_c) d\mathcal{Z}$. Based on this, we use a global self-adaptive graph $G_{\text{adp}}^{\text{mev}} \in \mathbb{R}^{c \times c}$ to capture these relatively stable relationships induced by the $\mathcal{Z}$.

Response 11:

Q11:- in Fig. 2, there are 2 disconnected components (top part (e), and another plot containing (a),(b),(c),(d),(e)). The notations should not be in the figure, but either already introduced in the text, or simplified to convey the main message. There are currently too many levels of blocks. On the pdf version, the inner blocks are not readable. Hard work has been put to make this plot, which can be useful for understanding the full details of the methodology, but for the article, it should be streamlined and connected to the main challenges. If possible, the top plot (e) can be reworked and put in Section 1, with highlight and the novelty and the capability offered by each new component (instead of notations difficult to grasp like AS-FFMLP, VT-ITGraph and FDG-Conv). - the same apply for Fig. 3.

Re: Dear Reviewer, thank you for your detailed comments regarding Fig.2. The caption for Fig.2 is intended to serve as a comprehensive introduction to the overall architecture of the VTITG-ASFFN model. We have integrated the caption from below

Fig.2 to serve as the introduction to the methodology section. This allows for a more natural transition into the subsequent descriptions of the VTITG-ASFFN model's core modules and its forward computation process. We have noticed the inconsistent subplot labeling that you pointed out. In the revised manuscript, we have corrected this by relabeling the top part as **(f)** and have renamed it to the more descriptive title: "Visualization of Time Series Transformations Process through FDM Block Embedding." The full names corresponding to the module acronyms (AS-FFMLP, VT-ITGraph, and FDG-Conv) can be found in the beginning of methodology. We also understand your desire for a more integrated connection between the figures and the paper's main challenges. While the page constraints made it difficult to move the diagram, we have revised part **(e)** within Fig. 2 to now explicitly state the main function of each module and the corresponding **Challenge (I, II, or III)** that it addresses.



(The VTITG-ASFFN architecture for historical and future MTS embedding learning.)

4 Methodology

In this section, we will delve into our proposed framework VTITG-ASFFN and core component modules. The VTITG-ASFFN architecture in Fig. 2 , which consists of History-Embedding *(a)*, Prediction *(b)* and Future-Embedding *(c)*. We introduce Adaptive-Smooth Frequency Fusion MLP (ASFF-MLP), Variable-Tailored Inter-Temporal Graph (VT-ITGraph), and Frequency Dilated Group Convolution (FDG-Conv) to create stackable Frequency Domain Multivariate Block (FDMB), fully leveraging them for end-to-end embedding learning. Furthermore, we use CI-MLP for predicting future time series based on historical data. The VTITG-ASFFN operates through process: ***History-Embedding*** $\rightarrow$ ***Prediction*** $\rightarrow$ ***Future-Embedding***.

Similarly, we understand the need to connect Fig. 3 to the methodological process of the Variable-Tailored Inter-Temporal Graph. We have now added annotations below the figure to clearly mark each stage of this process, which include: (1)

Variable Embeddings, (2) Targeted Adaptive Graph for MAV and MEV, (3) Transfer to Temporal Context, and (4) Capturing "Dynamic Spatial-Temporal Dependencies."

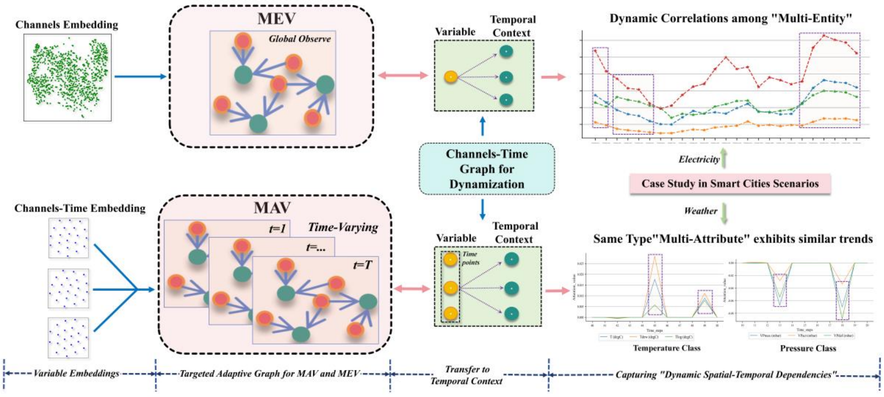


Fig. 3 Detailed layout of Variable-Tailored Inter-Temporal Graph. Top: VT-ITGraph for MEV task; Bottom: VT-ITGraph for MAV task. The VT-ITGraph consists of a tailored Adaptive Graph learner $G_{\text{adp}}^{\text{mav/mev}}$ and a Channels-Time Graph learner G_{CT}

Thank you for your rigorous feedback on our paper's figures! We see this as a sign of your appreciation for our work and your expectation for its further improvement.

Response 12:

Q12: Figs. 6, 7, 8 are difficult to understand: the amount of information in the plot (colors, highlighted components, annotated text, etc.) is disconnected to the legend of the figure. Once printed, it is not possible to make the difference between the different colors in the table.

Re: Dear Reviewer, thank you for your feedback on the figures in our Discussion section. We recognize that they may be difficult to grasp quickly for readers who have not yet delved deeply into the text. Considering the comprehensive nature of the experiments in this section (we conducted a total of nine different experiments to fully validate the feasibility of VTITG-ASFFN), splitting or further refining all figures would consume a significant amount of space. Therefore, we have done our best to convert some non-essential figures into a more compact table format. For example, the hyperparameter analysis is now presented in a simplified table instead of the original plot. Regarding the critical issue of distinguishing colors in printed versions, **we have taken your feedback seriously and revised the figures accordingly**. We agree that relying on color alone is insufficient. In the revised figures, we now use a combination of **different line styles and markers** in addition to a grayscale-friendly color palette. This ensures that all information remains clearly distinguishable even when the paper is printed in black and white.

Additionally, we have enlarged all figures and tables in the Discussion section as much as possible, while ensuring we remain within the 20-page limit. Regardless, we are very grateful for your high expectations for our paper and for providing such detailed suggestions for its improvement!

Technical questions

Response 13:

Q13:- I found the challenge 1 and the way to answer to it interesting. I have two questions/remarks: + The methodology is different based on whether the data is MAV or MEV. It seems that this MAV/MEV information is assumed to be known. Does it create a bias in favor of your methodology (since the other models use the same methodology for both datasets)? + How your methodology should be used when the data contain both MAV and MEV components?

Re: Dear Reviewer, we are delighted that you have raised such a technically insightful question. This is a point that we have also been deeply contemplating since our initial submission. We will provide a detailed analysis on these two points:

(1) You are correct. We agree that when not considering a mixed scenario, our variable-tailored inter-temporal graph does rely on prior domain knowledge, and an arbitrary choice could lead to suboptimal performance for VTITG-ASFFN. For example, if we were to apply the MAV-style time-varying graph to the large-scale Traffic dataset, the resulting graph would not only be enormous due to the large number of nodes, but it would also be prone to overfitting the shared temporal evolutionary trends on the training set (as suggested by the similarity observation for MEV in Fig. 1). This creates a fundamental mismatch, as our MAV model assumes each variable is governed by its own unique latent process, whereas our MEV assumption is that all entity variables are driven by a shared latent factor (e.g., the flow state of a regional traffic system). Therefore, for domain-specific datasets where the MAV or MEV characteristic is clear, we argue that one can rely on the inherent attribute names of the time series columns to determine whether to use the MAV-specific or MEV-specific version of VTITG-ASFFN. This frames the choice as an informed one based on available domain knowledge, rather than an unfair bias. Additionally, we provide a comparative experiment on the Weather, Finance, and Traffic datasets featuring three variants of VTITG-ASFFN. The findings reveal that the optimal performance of VTITG-ASFFN is achieved only when the specific variable-tailored inter-temporal graph is used for its corresponding scenarios:

Weather (MAV)	96		Finance (MIX)	12		Traffic (MEV)	96	
	MAE	MSE		MAE	MSE		MAE	MSE
VTITG-ASFFN(MAV)	**0.175**	**0.137**	VTITG-ASFFN(MAV)	0.476	0.703	VTITG-ASFFN(MAV)	0.407	0.758
VTITG-ASFFN(MEV)	0.182	0.143	VTITG-ASFFN(MEV)	0.355	0.444	VTITG-ASFFN(MEV)	**0.271**	**0.476**
VTITG-ASFFN(MIX)	0.185	0.148	VTITG-ASFFN(MIX)	**0.346**	**0.400**	VTITG-ASFFN(MIX)	0.279	0.494

(2) For datasets where MAV and MEV are mixed or difficult to distinguish beforehand, we propose in our revised manuscript the use of an **adaptive gating**

mechanism to dynamically fuse the MAV and MEV inter-temporal graphs. This results in a variable-tailored inter-temporal graph suitable for mixed MAV-MEV scenarios. You can find the technical details of this approach in the latter part of **Section 4.2** in our revised manuscript.

• *MAV-MEV Mixing:* Notably, while mainstream MTSF benchmarks can be categorized into attribute-dominant and entity-dominant types, hybrid scenarios with ambiguous feature representations commonly exist. To address this, we integrate a time-varying graph $G_{\text{adp}}^{\text{mav}} \in \mathbb{R}^{c \times c \times t}$ and a global enhanced graph $G_{\text{adp}}^{\text{mev}} \in \mathbb{R}^{c \times c}$ via adaptive gating fusion, where a dominance coefficient $\alpha_c \in \mathbb{R}^{c \times 1 \times 1}$ is learned to indicate feature tendency: $\alpha_c \to 1$ favors attribute features whereas $\alpha_c \to 0$ emphasizes entity characteristics. The resultant hybrid graph $G_{\text{adp}}^{\text{mix}} \in \mathbb{R}^{c \times c \times t}$ effectively handles mixed attributes and entities feature scenarios in discussion 6.2:

$$G_{\text{adp}}^{\text{mix}} = \text{Sigmoid}(\alpha_c) \cdot G_{\text{adp}}^{\text{mav}} + (1 - \text{Sigmoid}(\alpha_c)) \cdot G_{\text{adp}}^{\text{mev}} \tag{10}$$

here Sigmoid$(\cdot)$ is used to constrain the gating weights, while α_c will automatically find the optimal "identity" for each time series variable as the model minimizes the prediction loss. $G_{\text{adp}}^{\text{mev}}$ by broadcasting on the time-axis "t" to match $G_{\text{adp}}^{\text{mav}}$.

Channels-Time Graph for Dynamization. Owing to the dynamic nature of VT-ITGraph manifested in temporal evolution, we further augment it with a Channels-Time Graph $G_{\text{CT}} \in \mathbb{R}^{c \times t}$ to self-learning insights into the temporal variations of MTS, meanwhile the learning outcomes of Adaptive Graph from MAV/MEV is aggregated. Thus, we get transferred VT-ITGraph weight $W^{(\text{VT-ITG})}$ in temporal context:

$$P(G_{\text{CT}} \in \{\mathcal{A}\}) = \sum\nolimits_{X \in \{\mathcal{O}\}} P(G_{\text{adp}}, X) P(G_{\text{CT}} | G_{\text{adp}}, X), \tag{11}$$

$$W_{\text{mav}}^{(\text{VT-ITG})} = \sigma_{\text{gelu}}(X^{(\text{ASFF-MLP})} \otimes G_{\text{adp}}^{\text{mav}} \otimes G_{\text{CT}}), \quad \in \mathbb{R}^{b \times c \times t} \tag{12}$$

$$W_{\text{mev/mix}}^{(\text{VT-ITG})} = \sigma_{\text{tanh}}(X^{(\text{ASFF-MLP})} \otimes G_{\text{adp}}^{\text{mev/mix}} \otimes G_{\text{CT}}), \quad \in \mathbb{R}^{b \times t \times t} \tag{13}$$

$$X^{(\text{VT-ITG})} = X^{(\text{ASFF-MLP})} \otimes W_{\text{mav/mev/mix}}^{(\text{VT-ITG})} + X^{(\text{ASFF-MLP})}, \quad \in \mathbb{R}^{b \times c \times t} \tag{14}$$

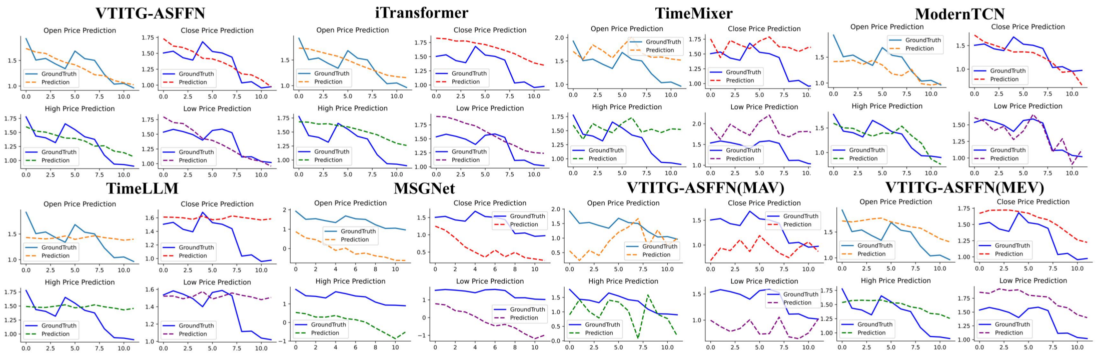
here $W^{(\text{VT-ITG})}$ in MAV task $\in \mathbb{R}^{b \times c \times t}$, "$c \times t$" indicates diverse attributes temporal trend; In MEV and MAV-MEV mixed task $\in \mathbb{R}^{b \times t \times t}$, "$t \times t$" means global temporal correlations. $\sigma(\cdot)$ denotes activation function (gelu or tanh) to enhance non-linear capability. G_{CT} essentially relies on the pre-learning of G_{adp} and X, but more importantly it establishes a transformation between multivariate correlations to temporal context, which leads to the mining for more explicit or latent temporal patterns.

To rigorously evaluate our model's performance in ambiguous scenarios, we introduce a new research question, **Discussion RQ2**, supported by a newly constructed synthetic **'Finance' dataset**. This dataset comprises time series from 20 representative leading stocks from the manufacturing and real economy sectors listed on the Shenzhen Stock Exchange. For each stock, we extracted four price attributes: open, close, high, and low, resulting in 80 time series variables. This construction deliberately creates a hybrid MAV-MEV scenario: any single variable can be interpreted both as an **'attribute'** (one of four price types) and as an **'entity'** (one of 20 companies), directly addressing the challenge of distinguishing between these characteristics. On this dataset, we conducted a comparative analysis of our VTITG-ASFFN variant designed for mixed scenarios against leading state-of-the-art (SOTA) baselines from diverse architectural categories, including iTransformer (Transformer-based), TimeMixer (MLP-based), ModernTCN (TCN-based), Time-LLM (LLM-based), and MSGNet (Graph-based).

Table 4 MAV-MEV mixed comparisons in the synthetic "Finance" dataset (input-12-predict-12).

Model	Ours (MIX)		Ours (MAV)		Ours (MEV)		iTransformer		TimeMixer		ModernTCN		Time-LLM		MSGNet	
Metric	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Finance	**0.346**	**0.400**	0.476	0.703	0.355	0.444	<u>0.347</u>	<u>0.404</u>	0.350	0.413	0.367	0.462	0.355	0.419	0.422	0.546

The results validate the effectiveness of our approach. Furthermore, a visualization of the predictions confirms that the **VTITG-ASFFN(MIX)** variant excels at fitting the overall trend, significantly outperforming the single-logic MAV-only and MEV-only variants. In contrast, the baseline models exhibit distinct weaknesses: the predictions from TimeMixer and TimeLLM deviate significantly from the ground truth; while iTransformer and MSGNet capture the general trend, their prediction curves show a poor fit; and ModernTCN is susceptible to distribution shift and exhibits a noticeable prediction lag. These findings validate our central hypothesis: in scenarios where pre-defining variables as either MAV or MEV is difficult, our proposed design using **adaptive gating to fuse different variable graphs offers a distinct advantage**. Please note that while this visualization was generated for analysis, we were unable to include it in the revised manuscript due to the 20-page limit.



(Prediction Test of VTITG-ASFFN on Synthetic Dataset "Finance": input-12-predict-12. The blue line represents the actual values, while the orange/red/green/purple line depicts the predicted values. All time series data have been standardized)

Response 14:

Typos

hadamard --> Hadamard
xavier --> Xavier

In Section 1: "prior knowledges", "fails", "tends"
In Section 6: "extractd"
T-SNE --> t-SNE

Re: Dear Reviewer, thank you for your meticulous work in identifying the typos in our manuscript. We are very grateful for the close attention and detailed reading you have given our paper. We have corrected all of the typos you pointed out in the revised version of the manuscript. Thank you again for your detailed review. We have done our best to address all of your revision requests and hope that we have comprehensively resolved your concerns.

To Reviewer 3：

Response 1:

Q1: The abstract places notable emphasis on smart city applications, which may lead readers to expect an in-depth analysis or case study specifically within that context. However, the main body of the paper does not provide sufficient discussion or validation tied to smart city scenarios, instead only using several datasets from these scenarios, which makes this framing appear somewhat misleading. I recommend either providing the smart city relevance or revising the abstract to more accurately reflect the paper’s scope.

Re: Dear Reviewer, we thank you for your insightful comment regarding the introduction of the "smart cities" concept in our abstract. We agree with your assessment and have come to realize that our VTITG-ASFFN model is primarily oriented towards the general-purpose task of multivariate time series forecasting. Forcing the "smart cities" application context onto this foundation could detract from the paper's core focus on our methodological improvements for MTSF. Accordingly, we have revised the manuscript to remove the vast majority of references to "smart cities". The concept is now mentioned only in the dataset description section to provide context, indicating that the datasets originate from real-world smart city applications. We are very grateful for your high-level, structurally guiding feedback. It has been instrumental in helping us to better delimit the research scope of our paper.

Response 2:

Q2: It remains unclear whether the two FDM blocks used in the model are trained independently or share weights becase the function name are the same in Fig. 1 as well as in Equation (20) and (22). This architectural detail is essential for reproducibility and understanding model complexity. The paper should explicitly describe how the two blocks interact, whether they are identical in design, and if any parameter sharing or joint optimization is employed.

Re: Dear Reviewer, thank you for pointing out the lack of clarity regarding Equations (20, 22). The two FDMBs serve as the concrete implementations for the "History-Embedding" and "Future-Embedding" stages, respectively. Their parameters are independent (i.e., not shared) and are both optimized through a supervised, error-driven gradient descent process. The purpose of the first FDMB ('History-Embedding') is to perform Frequency Domain Multivariate learning on the input time series. In contrast, the purpose of the second FDMB ('Future-Embedding') is to operate on the time series generated by the 'Prediction' stage to correct the discrepancy between the predictive and true distributions. The difference between "History-Embedding" and "Future-Embedding" also lies in the length of the time series they process (i for the historical sequence and o for the future forecast sequence). Thank you again for this valuable suggestion. We have revised the descriptions and notation in the FDMB section, as well as the description of the VTITG-ASFFN's training mechanism, to explicitly clarify the distinct roles of "History-Embedding" and "Future-Embedding" in the manuscript.

4.4 Model Formulation and Forward Calculation

The Frequency Domain Multivariate Block implements the sequential encapsulation of the modules presented in the previous section 4.1, 4.2, 4.3:

$$X^{(\text{ASFF-MLP})} = \text{ASFF-MLP}(X), \tag{17}$$

$$W^{(\text{VT-ITG})} = \text{VT-ITGraph}(X^{(\text{ASFF-MLP})}), \tag{18}$$

$$X^{(\text{VT-ITG})} = X^{(\text{ASFF-MLP})} \times \sigma(W^{(\text{VT-ITG})}) + X^{(\text{ASFF-MLP})}, \tag{19}$$

$$X^{(\text{FDG})} = \text{FDG-Conv}(X^{(\text{VT-ITG})}). \tag{20}$$

where $X^{(\text{ASFF-MLP})}$, $X^{(\text{VT-ITG})}$, $X^{(\text{FDG})}$ all $\in \mathcal{R}^{b \times c \times t}$. Finally for calculation, we regard the forward process of VTITG-ASFFN as an end-to-end learning in Fig. 2:

- *History Embedding:* Learn the historical spatio-temporal correlations through FDMB$(\cdot)$ stacking: $X^{(l)} = \text{FDMB}_{\text{his}}^{(l)}\left(X^{(l-1)}\right), l = 2, ..., L$. Where $X^{(l)}$ all $\in \mathcal{R}^{b \times c \times i}$.
- *Prediction:* Use the CI-MLP$(\cdot)$ to predict $Y \in \mathcal{R}^{b \times c \times o}$: $Y = \text{CI-MLP}(X^{(\text{L})})$.
- *Future Embedding:* Perform the same operation on Y as History Embedding to get the final MTSF result $\hat{X}$: $Y^{(l)} = \text{FDMB}_{\text{pred}}^{(l)}\left(Y^{(l-1)}\right), l = 2, ..., L$. Where $Y^{(l)}$ all $\in \mathcal{R}^{b \times c \times o}$, $\hat{X} = Y^{(L)}$ and $\text{FDMB}_{\text{his}}^{(l)}$, $\text{FDMB}_{\text{pred}}^{(l)}$ are parameter-independent.

Noted that the objective of the Future-Embedding is to minimize the distance between the predictive distribution and the true distribution due to the label auto-correlation and temporal distribution shift of future time series. Meanwhile we choose $L1$ loss for VTITG-ASFFN's training because of its robust to noises Han et al. (2024).

Response 3:

Q3 : The proposed distinction between MAV and MEV is insightful. However, the paper lacks a clear explanation of how this classification is operationalized in practice. What is the procedure used to determine whether a variable falls into WAV or WEV? Given the potential ambiguity in real-world datasets, such as certain subsets in the M4 dataset where variable type is not easily distinguishable, this kind of classification might limit model generalizability. Further elaboration or a robustness discussion would strengthen this contribution.

Re: Dear Reviewer, thank you for your insightful question. We will address this in two parts:

(1) Regarding the operationalization of MAV/MEV classification for the datasets in our original study, we note that most standard, general-purpose multivariate time series datasets have clear domain descriptions. Before training a domain-specific model, we can ascertain the physical meaning represented by each time series column. For example, the Weather dataset includes attributes such as atmospheric pressure, temperature, and wind speed, while the Traffic dataset contains entities (different

sensors recording traffic flow). Therefore, based on this prior domain knowledge, we can generally determine whether a given dataset is predominantly composed of multi-attribute or multi-entity variables before training begins.

(2) We completely agree with your assessment that real-world scenarios can present ambiguity where it is difficult to strictly demarcate between 'multi-attribute' and 'multi-entity' types. The M4 dataset is an excellent example of this, as it covers a wide range of application domains. However, we found that the high degree of domain mixing in M4 might not be the ideal setting to clearly evaluate how VTITG-ASFFN performs in a hybrid MAV and MEV scenario.

Therefore, to directly address your point, we constructed a new synthetic dataset specifically for this purpose, using the stock market as an example. We selected 20 representative leading stocks from the manufacturing and real economy sectors listed on the Shenzhen Stock Exchange. For each stock, we extracted four price behavior attributes: open, close, high, and low, resulting in a total of 80 time series variables. In this setup, any single time series variable can be viewed from two perspectives: it belongs to an **attribute** category (open, close, high, or low price) and simultaneously belongs to an **entity** category (one of the 20 listed companies). This creates the exact hybrid MAV-MEV scenario you described.

To handle this, we have adopted a **Variable-Tailored Inter-Temporal Graph based on adaptive gating fusion**, and we have added a description of this design to **Section 4.2** of our methodology to specifically address such mixed MAV-MEV cases.

- *MAV-MEV Mixing:* Notably, while mainstream MTSF benchmarks can be categorized into attribute-dominant and entity-dominant types, hybrid scenarios with ambiguous feature representations commonly exist. To address this, we integrate a time-varying graph $G_{\text{adp}}^{\text{mav}} \in \mathbb{R}^{c \times c \times t}$ and a global enhanced graph $G_{\text{adp}}^{\text{mev}} \in \mathbb{R}^{c \times c}$ via adaptive gating fusion, where a dominance coefficient $\alpha_c \in \mathbb{R}^{c \times 1 \times 1}$ is learned to indicate feature tendency: $\alpha_c \to 1$ favors attribute features whereas $\alpha_c \to 0$ emphasizes entity characteristics. The resultant hybrid graph $G_{\text{adp}}^{\text{mix}} \in \mathbb{R}^{c \times c \times t}$ effectively handles mixed attributes and entities feature scenarios in discussion 6.2:

$$G_{\text{adp}}^{\text{mix}} = \text{Sigmoid}(\alpha_c) \cdot G_{\text{adp}}^{\text{mav}} + (1 - \text{Sigmoid}(\alpha_c)) \cdot G_{\text{adp}}^{\text{mev}} \tag{10}$$

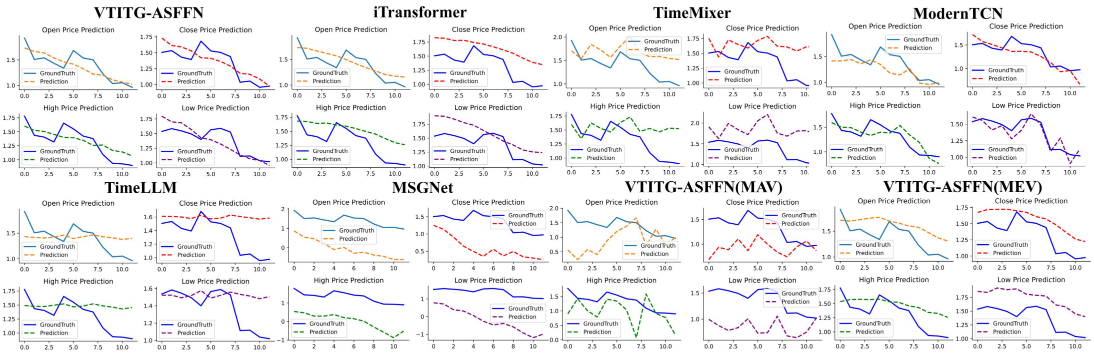
here $\text{Sigmoid}(\cdot)$ is used to constrain the gating weights, while α_c will automatically find the optimal "identity" for each time series variable as the model minimizes the prediction loss. $G_{\text{adp}}^{\text{mev}}$ by broadcasting on the time-axis "t" to match $G_{\text{adp}}^{\text{mav}}$.

Subsequently, in a new **Discussion RQ2**, we introduce the synthetic 'Finance' dataset that we constructed. On this dataset, we conduct a comparative analysis against leading state-of-the-art (SOTA) baselines from diverse architectural categories: iTransformer (Transformer-based), TimeMixer (MLP-based), ModernTCN (TCN-based), Time-LLM (LLM-based), and MSGNet (Graph-based). This experiment demonstrates the effectiveness of our VTITG-ASFFN variant designed for mixed MAV-MEV scenarios, particularly in situations where the distinction between 'multi-attribute' and 'multi-entity' characteristics is ambiguous.

Table 4 MAV-MEV mixed comparisons in the synthetic "Finance" dataset (input-12-predict-12).

Model	Ours (MIX)		Ours (MAV)		Ours (MEV)		iTransformer		TimeMixer		ModernTCN		Time-LLM		MSGNet	
Metric	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Finance	**0.346**	**0.400**	0.476	0.703	0.355	0.444	<u>0.347</u>	<u>0.404</u>	0.350	0.413	0.367	0.462	0.355	0.419	0.422	0.546

Additionally, we have provided a visualization of the actual predictions. This shows that the VTITG-ASFFN(MIX) variant also demonstrates excellent performance in fitting the overall trend, significantly outperforming the single-logic variants that use only the multi-attribute or multi-entity approach. In contrast, the baseline models exhibit issues to varying degrees: the predictions from TimeMixer and TimeLLM deviate significantly from the ground truth; while iTransformer and MSGNet capture the general trend, their prediction curves show a poor fit; and ModernTCN is susceptible to distribution shift and exhibits a noticeable prediction lag. This finding indicates that in scenarios where it is difficult to pre-define variables as either MAV or MEV, the design using adaptive gating to fuse different variable graphs offers a distinct advantage. However, due to the 20-page limit, we have not included this visualization in the revised manuscript.



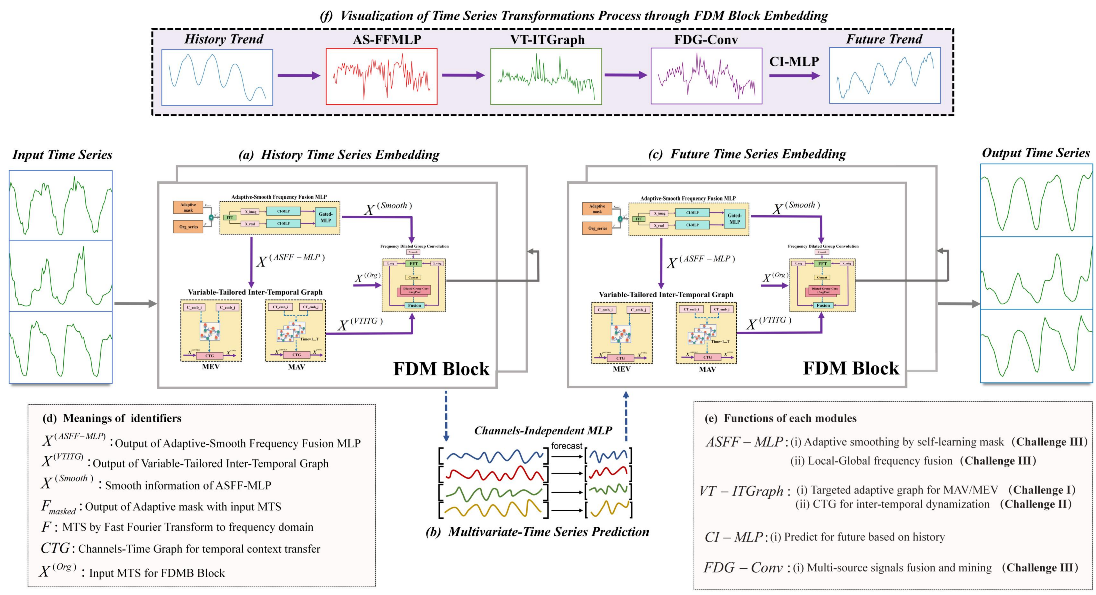
(Prediction Test of VTITG-ASFFN on Synthetic Dataset "Finance": input-12-predict-12. The blue line represents the actual values, while the orange/red/green/purple line depicts the predicted values. All time series data have been standardized)

Thank you again for such constructive feedback!

Response 4:

Q4: Fig. 2 (FDM block) is visually too small and difficult to interpret. It would be more effective to replace the detailed diagram in the main framework with a simple placeholder or abbreviation, and then provide an enlarged standalone illustration of the FDM block for clarity. Moreovere, the font size in figures and tables throughout the experimental section is generally too small, which hinders readability. I recommend increasing font size for improved presentation quality.

Re: Dear Reviewer, thank you for your valuable suggestions regarding the revision of our figures and tables. We acknowledge that the FDM block within Fig. 2 may still not be sufficiently large. However, we would like to respectfully justify our decision to present it within the main architectural diagram. The VTITG-ASFFN model is designed as an end-to-end pipeline that follows the **History-Embedding → Prediction → Future-Embedding** process. We believe a separate, detailed figure for the FDMB might be redundant because our current Fig. 2 is intended to illustrate not only the specific composition of the components within the FDMB, but also how they fit into the complete end-to-end workflow (a high-level summary of which is now also provided in the revised Fig. 2(f)). To complement the visual representation, we have also detailed the operational flow of both the FDMB and the overall architecture with equations in Section 4.4 (as mentioned in our Response 2). At the same time, we also recognize that the font size in our figures and tables is generally small. We did not intentionally set a small font size; this may be an artifact of the current template's page limits and style constraints. We will do our best to increase the font size and clarity of all figures and tables as much as possible in the final camera-ready version. Furthermore, in the revised manuscript, we have also converted some figures into tables to improve clarity, for instance, by using a table for the hyperparameter analysis instead of the original figure. Thank you again for your valuable suggestions regarding the revision of our paper's figures and tables. Taking all of your comments into consideration, we have comprehensively optimized all figures and tables in the paper to significantly improve their clarity, readability, and overall presentation quality, while ensuring that no essential content was sacrificed.



(The VTITG-ASFFN architecture for historical and future MTS embedding learning.)

To Reviewer 4：

Response 1:

Q1: Ablation Studies: To strengthen the robustness of the findings, the authors should conduct ablation experiments on datasets with longer time steps (e.g., ETT and DSMT). This would demonstrate the model's efficacy in scenarios with complex temporal dynamics.

Re: Dear Reviewer, thank you for your valuable suggestions. Our original ablation study comprehensively considers forecast horizons ranging from short to long-term ({96, 192, 336, 720}), where the 720-step horizon already represents a long-term span that includes sufficient temporal complexity and multivariate dynamics. Due to space constraints, we only presented the averaged results in the initial manuscript. We have now provided the full numerical results for the ablation study across all forecast horizons ({96, 192, 336, 720}):

Datasets	Strategy	96		192		336		720	
		MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE
Weather	Benchmark	0.180	0.142	0.226	0.186	0.268	0.232	0.308	0.291
	wo/ASFF-MLP	0.194	0.151	0.247	0.206	0.283	0.255	0.329	0.317
	wo/w-MLP	0.195	0.153	0.237	0.196	0.277	0.243	0.323	0.306
	wo/VT-ITGraph	0.202	0.162	0.242	0.205	0.281	0.255	0.340	0.331
	wo/w-ADPGraph	0.185	0.146	0.231	0.192	0.271	0.242	0.318	0.301
	wo/FDG-Conv	0.278	0.201	0.312	0.241	0.346	0.282	0.375	0.326
	wo/w-Conv	0.184	0.144	0.229	0.189	0.272	0.237	0.339	0.328
	wo/Future-Embedding	0.182	0.146	0.232	0.194	0.272	0.242	0.318	0.303
Traffic	Benchmark	0.271	0.475	0.277	0.481	0.282	0.493	0.303	0.540
	wo/ASFF-MLP	0.343	0.608	0.319	0.565	0.335	0.608	0.352	0.647
	wo/w-MLP	0.297	0.519	0.296	0.520	0.302	0.541	0.322	0.585
	wo/VT-ITGraph	0.328	0.594	0.327	0.582	0.331	0.589	0.351	0.628
	wo/w-ADPGraph	0.293	0.512	0.289	0.501	0.297	0.517	0.318	0.56
	wo/FDG-Conv	0.585	1.166	0.413	0.763	0.409	0.756	0.414	0.765
	wo/w-Conv	0.320	0.624	0.333	0.644	0.345	0.661	0.325	0.620
	wo/Future-Embedding	0.278	0.487	0.301	0.521	0.307	0.534	0.327	0.574

For clarity, the notation used is as follows: "wo/(ASFF-MLP/VT-ITGraph/FDG-Conv/Future-Embedding)" signifies that the outputs of these components are zeroed out to isolate their contributions, while "wo/w-(MLP/ADPGraph/Conv)" indicates the replacement of ASFF-MLP/VT-ITGraph/FDG-Conv with a standard MLP/Adaptive Graph/Convolution. These more granular, per-horizon results confirm that our proposed components—the Adaptive-Smooth Frequency Fusion MLP (ASFF-MLP), the Variable-Tailored Inter-Temporal Graph (VT-ITGraph), and the Frequency Dilated Group Convolution (FDG-Conv)—are all effective.

Of course, we also deeply recognize the importance of your suggestion to conduct ablations in more complex time series scenarios. We believe the ETT and DSMT datasets are indeed complex, as they correspond to the real-world dynamics of energy utilization and a complex remote sensing system, respectively. Therefore, we have now conducted separate ablation studies on both of these datasets. Notably, for the DSMT dataset, we performed ablations on horizons of {512, 1024, 2048, 3072} steps to evaluate the performance changes of VTITG-ASFFN after removing key modules in an ultra-long-term forecasting setting. The experiments prove that the main components of our proposed VTITG-ASFFN are indeed effective in both complex and ultra-long-term prediction scenarios. In particular, the results show that in the ultra-long-term setting, the ASFF-MLP, VT-ITGraph, and FDG-Conv all play a significant role in the model's performance. This is because they are responsible for **temporal learning**, **multivariate dependencies**, and **fusing multi-source signals and mining primary temporal patterns**, respectively. These functions become increasingly crucial in ultra-long-term forecasting, where the time series distribution drift becomes more pronounced with larger look-back windows.

ETTm2	96		192		336		720		AVERAGE			
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	OWA	
Benchmark	0.218	0.111	0.250	0.142	0.277	0.173	0.314	0.218	0.26467	0.16085	0.21276	
wo/ASFF-MLP	0.225	0.115	0.256	0.146	0.290	0.188	0.330	0.239	0.27525	0.172	0.223625	0.05106693
wo/VT-ITGraph	0.223	0.116	0.251	0.144	0.283	0.181	0.331	0.239	0.272	0.17	0.221	0.038729084
wo/FDG-Conv	0.378	0.271	0.402	0.303	0.427	0.334	0.450	0.369	0.41425	0.31925	0.36675	0.723773266

DSMT-D1	96		192		336		720		AVERAGE			
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	OWA	
Benchmark	0. 061	0. 017	0. 121	0. 056	0. 180	0. 104	0. 157	0. 092	0.12975	0.06725	0.0985	
wo/ASFF-MLP	0. 073	0. 023	0. 135	0. 068	0. 200	0. 132	0. 290	0. 240	0.1745	0.11575	0.145125	0. 473350254
wo/VT-ITGraph	0. 067	0. 021	0. 129	0. 067	0. 191	0. 122	0. 267	0. 203	0.1635	0.10325	0.133375	0. 354060914
wo/FDG-Conv	0. 307	0. 163	0. 346	0. 195	0. 385	0. 232	0. 441	0. 305	0.36975	0.22375	0.29675	2. 012690355

DSMT	512		1024		2048		3072		AVERAGE			
	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	OWA	
Benchmark	0.199	0.125	0.258	0.191	0.307	0.249	0.332	0.281	0.274	0.2115	0.24275	
wo/ASFF-MLP	0.702	0.830	0.710	0.846	0.711	0.844	0.719	0.868	0.7105	0.847	0.77875	2.208032956
wo/VT-ITGraph	0.503	0.540	0.510	0.564	0.513	0.577	0.520	0.586	0.5115	0.56675	0.539125	1.220906282
wo/FDG-Conv	0.484	0.512	0.500	0.533	0.522	0.571	0.528	0.572	0.5085	0.547	0.52775	1.174047374

Response 2:

Q2: Computational Efficiency: The adaptive graph design for MAV (Multi-Attribute Variables) and MEV (Multi-Entity Variables) may introduce additional computational overhead. The authors should include a complexity analysis or compare computational costs with baseline methods to justify the trade-off between performance gains and efficiency.

Re: Dear Reviewer, thank you for your insightful suggestion. In fact, in Section 6.6 (RQ6) of our Discussion, we have already discussed the comparison of computational costs between VTITG-ASFFN and baseline methods in ultra-long-term forecasting. As shown, VTITG-ASFFN consistently maintains high computational efficiency. This efficiency is attributable to several key design choices: the **low-rank embedding dimension** of the adaptive graph and the **Frequency Dilated Group Convolution**, which is based on dilation factors and causal convolutions. These components significantly reduce the memory footprint (VRAM usage) of VTITG-ASFFN, and the convolutions also facilitate GPU acceleration for enhanced parallel computation efficiency. Furthermore, our **Adaptive-Smooth Frequency Fusion MLP** does not perform high-dimensional embedding learning, which avoids the introduction of an excessive number of high-dimensional parameters. We invite you to pay special attention to the "Memory.avg (MB)" and "Speed.avg (s/iter)" columns in Table 4 of our revised manuscript for a direct comparison of these metrics:

Table 4 Ultra-Long MTSF and Operational Efficiency in DSMT (512 input length). VTITG-ASFFN is efficient due to low embed dim VT-ITGraph and parallel FDG-Conv

Horizons	512		1024		2048		3072		Average		Memory.avg	Speed.avg
Models	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	(MB)	(s/iter)
VTITG-ASFFN*	**0.199**	**0.125**	**0.258**	**0.191**	**0.307**	**0.249**	**0.332**	**0.281**	**0.274**	**0.211**	**886**	**0.0024**
VTITG-ASFFN	**0.202**	**0.129**	**0.259**	**0.191**	**0.307**	**0.249**	**0.334**	**0.282**	**0.276**	**0.213**	3090	**0.0067**
iTransformer	0.217	0.147	0.277	0.220	0.338	0.301	0.370	0.342	0.301	0.253	**1582**	**0.0036**
TimeMixer	0.216	0.142	0.282	**0.216**	**0.335**	**0.279**	0.357	**0.311**	0.298	0.237	2340	0.0110
ModernTCN	0.218	0.154	0.283	0.239	0.345	0.321	0.380	0.368	0.307	0.271	**1230**	0.0074
MSGNet	0.230	0.163	0.295	0.249	0.359	0.338	0.400	0.400	0.321	0.288	2952	0.0498
Time-LLM	**0.206**	**0.134**	**0.267**	**0.203**	**0.323**	**0.274**	**0.356**	0.314	**0.288**	**0.231**	16036	0.2135

We agree that a discussion of complexity analysis is an excellent suggestion. However, we believe that in the field of multivariate time series forecasting, a purely theoretical complexity analysis may be insufficient to capture the full picture. In comparison, while many current state-of-the-art baseline models—based on Transformer, MLP, CNN, LSTM, GNN, or LLM architectures—employ various theoretical methods to reduce computational complexity, their actual resource consumption remains substantial. This is often due to the introduction of numerous practical implementation techniques, such as high-dimensional embedding enhancement (e.g., as seen in DSformer: A Double Sampling Transformer for Multivariate Time Series Long-term Prediction, CIKM 2023). Therefore, in our manuscript, we have only briefly mentioned the time and space complexity of the core components within VTITG-ASFFN, rather than engaging in an exhaustive theoretical discussion. Our primary objective is to demonstrate the practical efficiency of VTITG-ASFFN through empirical results, rather than focusing on an overly theoretical analysis.

Finally, regarding the trade-off between performance and efficiency, we have also analyzed this in the ultra-long-term scenario in Discussion RQ6. Using VTITG-ASFFN as an example, the VTITG-ASFFN* variant (which indicates the model without Future-Embedding) achieves better performance, a lower memory footprint, and faster computational speed than the native VTITG-ASFFN in Table 4. This is because the number of parameters in the full VTITG-ASFFN, which follows the History-Embedding → Prediction → Future-Embedding pipeline, grows rapidly with the prediction length. This is closely related to its end-to-end architecture, as we also apply FDMB embedding learning to the multivariate time series output from the "Prediction" stage. As the effects of distribution drift and the heterogeneity of ultra-long-term MTS become more significant, the latent temporal patterns mined by the Future-Embedding module become unreliable (as evidenced by the ineffective results in the second row of Table 4). Therefore, we removed the Future-Embedding module to avoid error accumulation and the additional costs associated with ultra-long-term forecasting. This implies that one must judiciously apply the stacked FDMB for embedding learning to strike a balance between predictive performance and computational overhead.

Finally, we would like to thank you again for your appreciation of our work and for your insightful feedback!